

An Electron-Hole Rich Dual-Site Nickel Catalyst for Efficient Photocatalytic Overall Water Splitting

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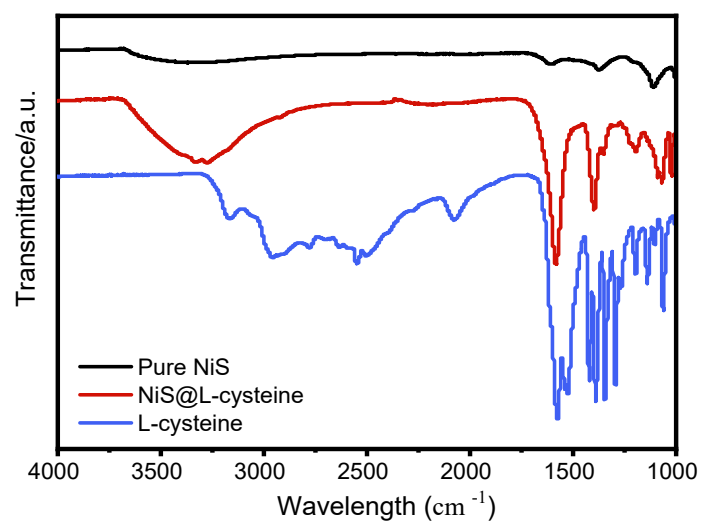
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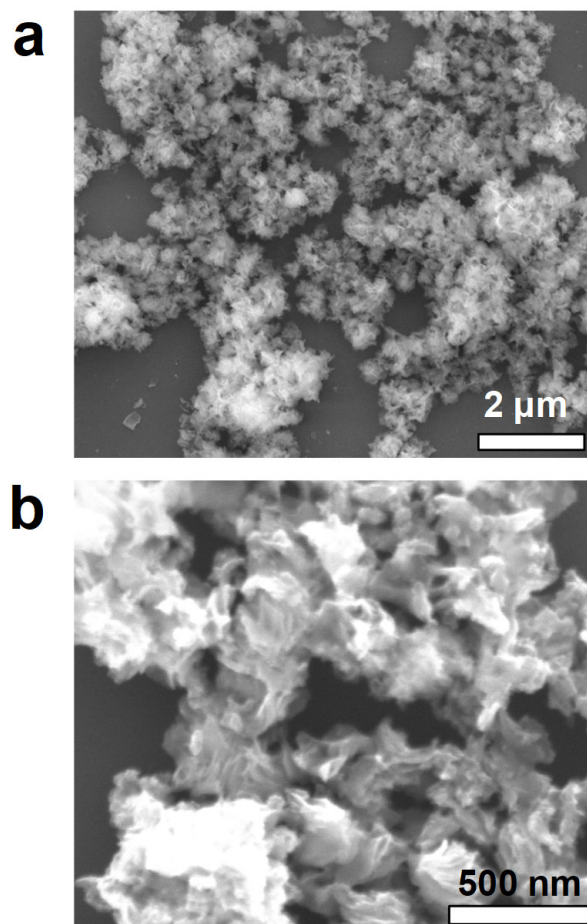
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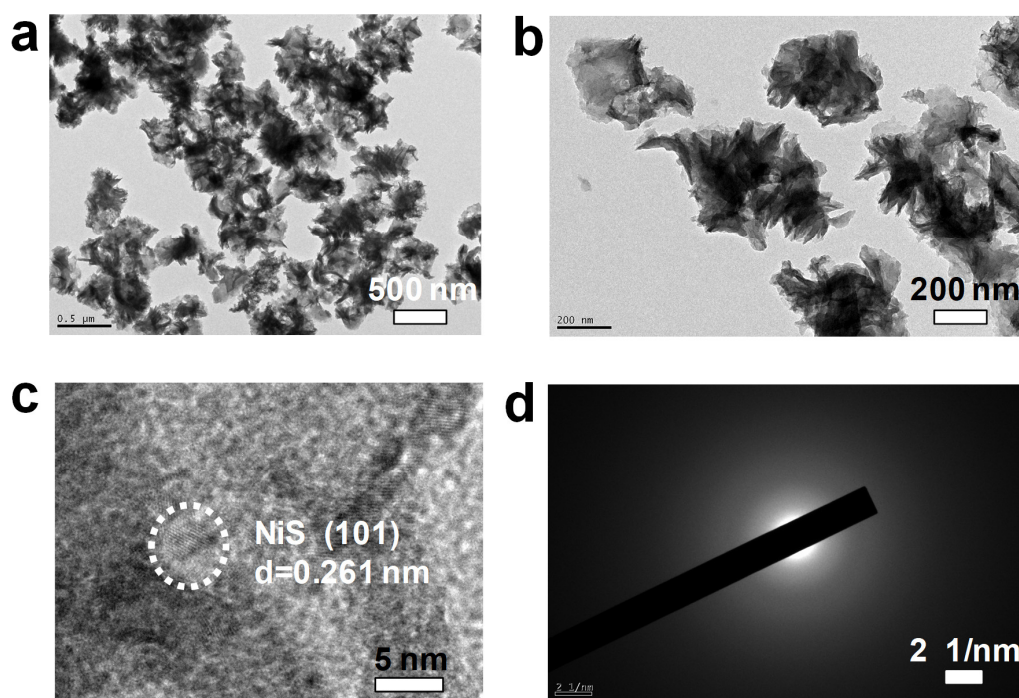
Supplementary References



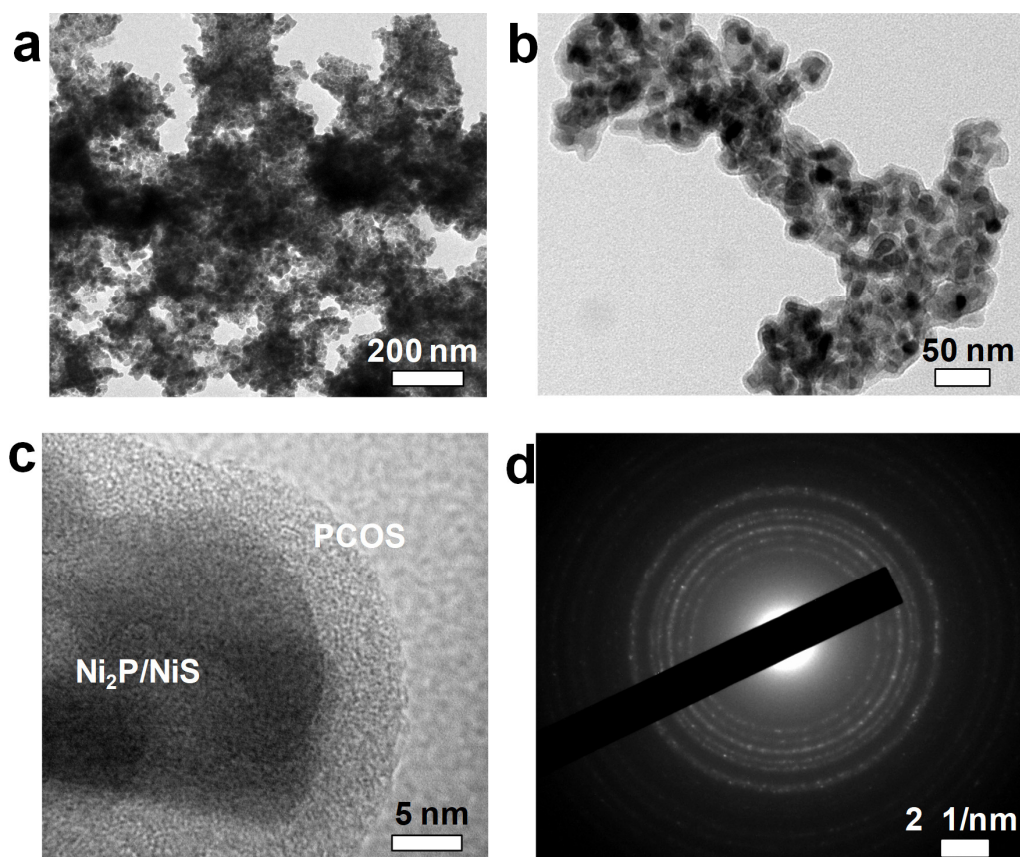
Supplementary Figure 1. FT-IR spectra of pure NiS nanosheets, NiS@L-cysteine and pure L-cysteine.



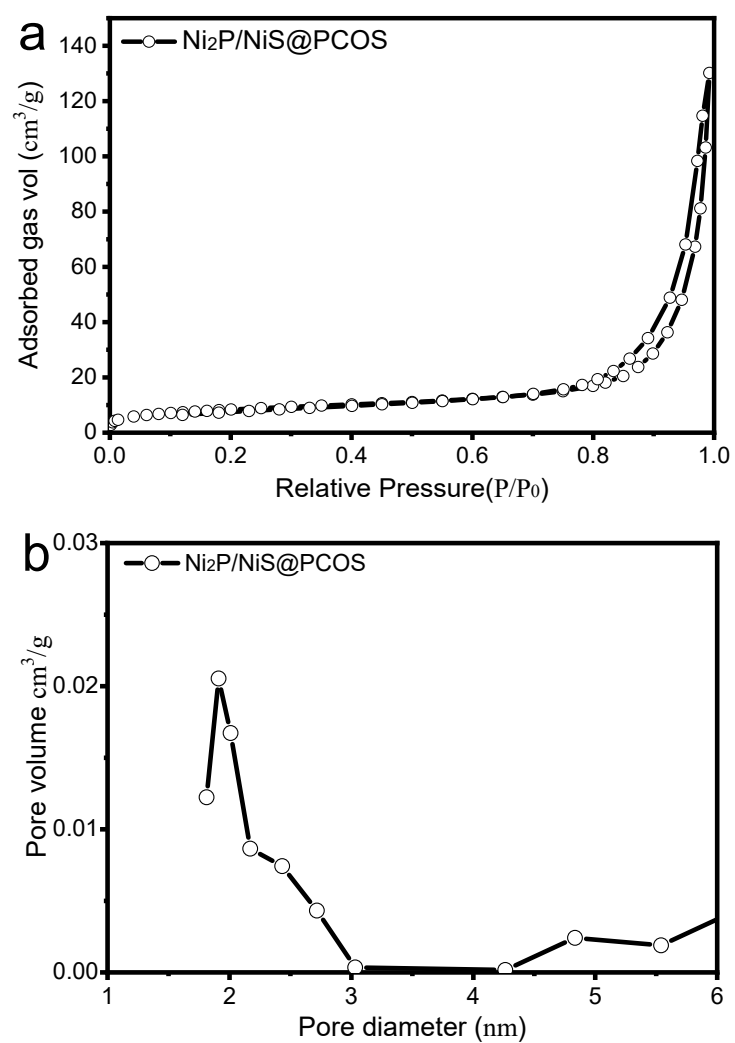
Supplementary Figure 2. (a, b) SEM images of the NiS@ L-cysteine.



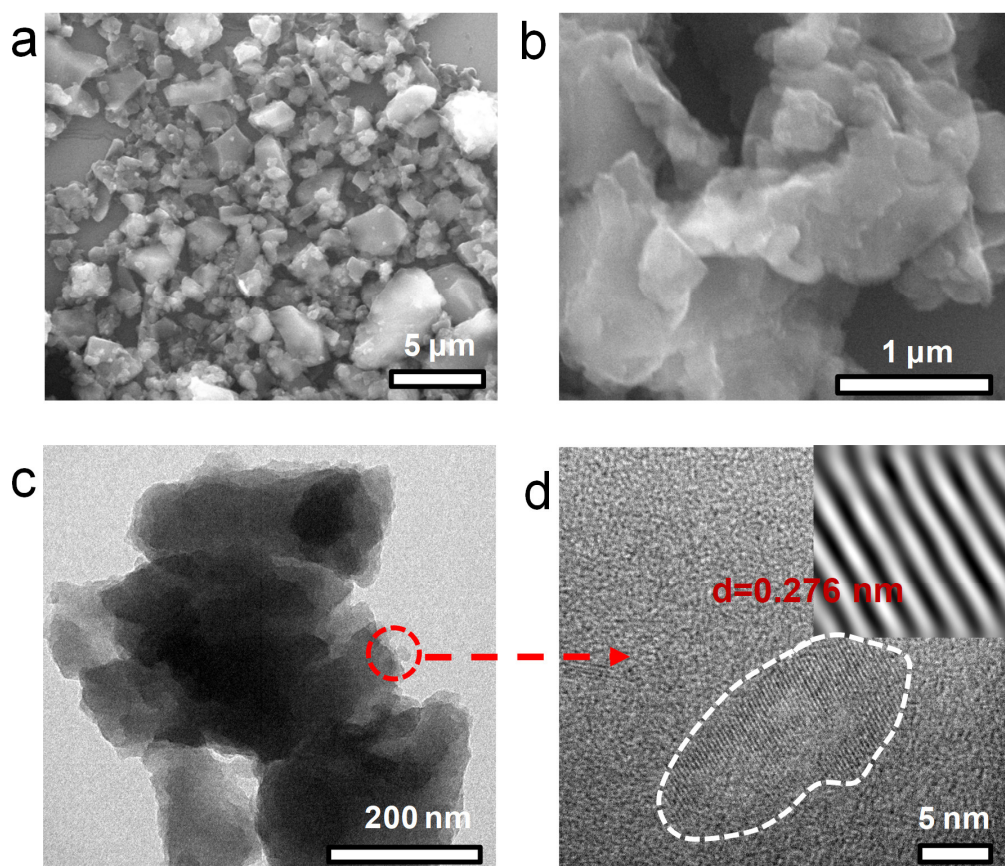
Supplementary Figure 3. TEM(a, b), HRTEM (c) and the selected-area electron diffraction (d) images of the NiS @L-cysteine.



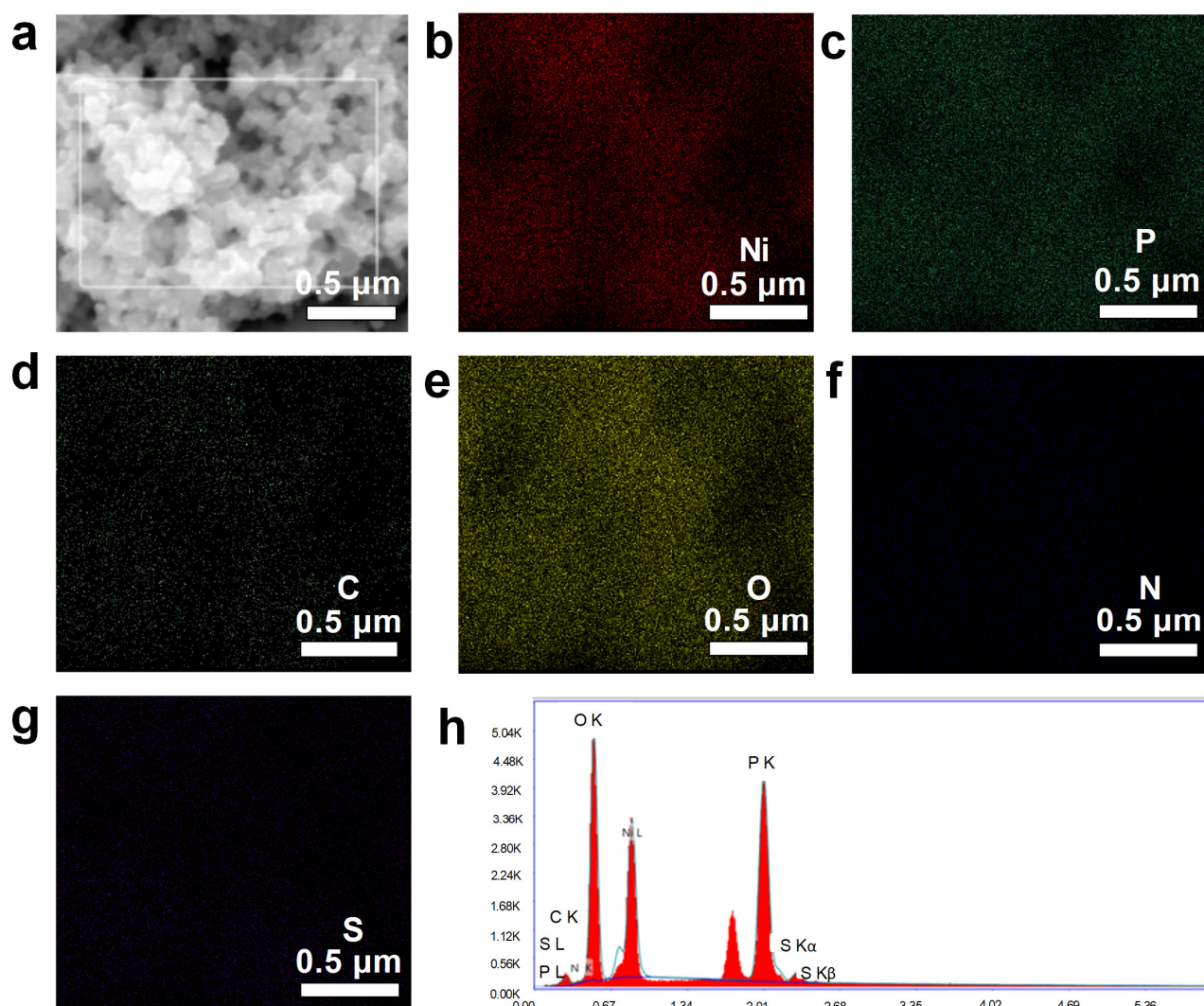
Supplementary Figure 4. (a-c) TEM images of the $\text{Ni}_2\text{P}/\text{NiS}@PCOS$. (d) the selected-area electron diffraction pattern of $\text{Ni}_2\text{P}/\text{NiS}@PCOS$.



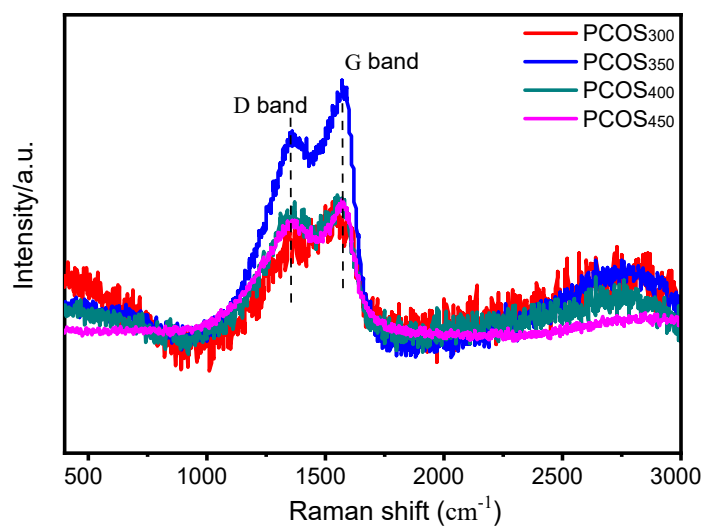
Supplementary Figure 5. (a) Nitrogen adsorption-desorption isotherms and (b) the corresponding pore size distributions curve of $\text{Ni}_2\text{P}/\text{NiS}@PCOS$ sample.



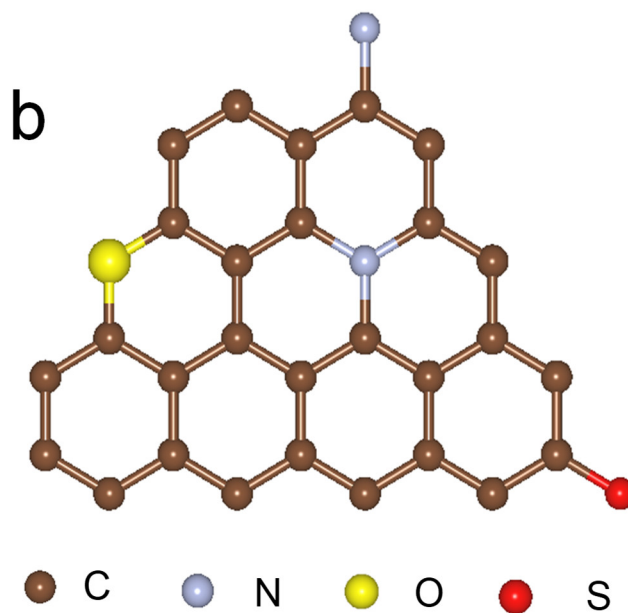
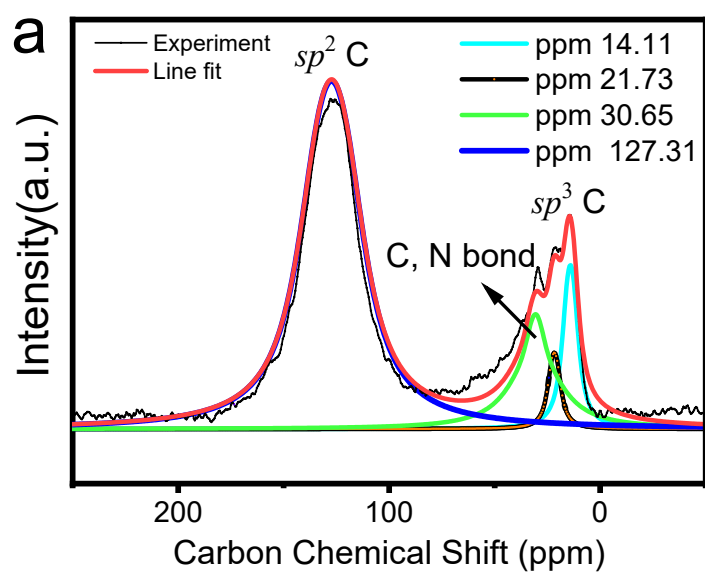
Supplementary Figure 6. (a-b) SEM and (c-d) TEM/HRTEM images of the polymeric carbon-oxygen semiconductor (PCOS₃₅₀).



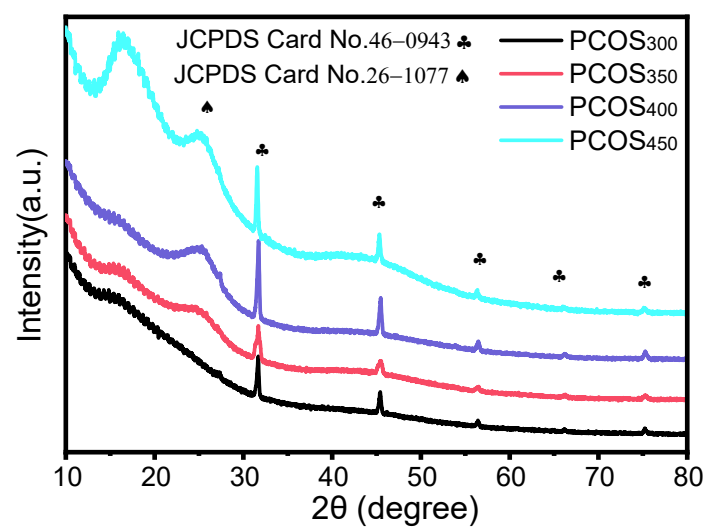
Supplementary Figure 7. The corresponding EDX elemental mapping images of $\text{Ni}_2\text{P}/\text{NiS}@/\text{PCOS}$ sample: (a) SEM images; (b) nickel (red); (c) phosphorus (green); (d) carbon (aqua); (e) oxygen (yellow); (f) nitrogen (blue); (g) sulfur (violet) in the selected area, respectively.



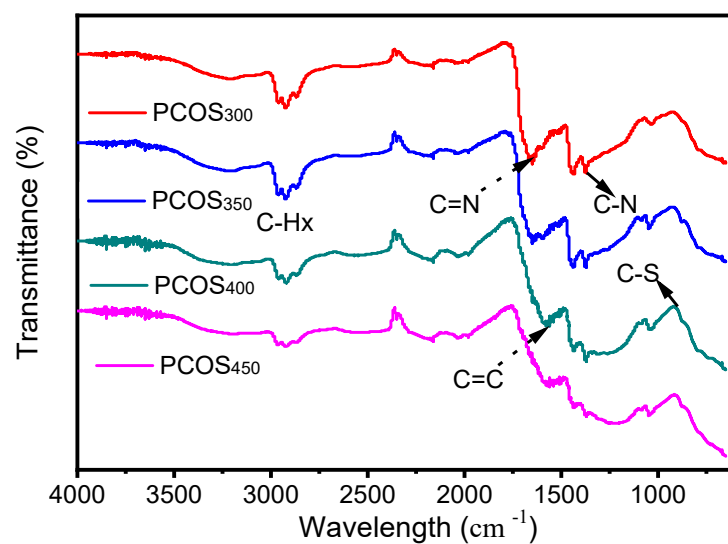
Supplementary Figure 8. Raman spectra of PCOS₃₀₀, PCOS₃₅₀, PCOS₄₀₀, PCOS₄₅₀, respectively.



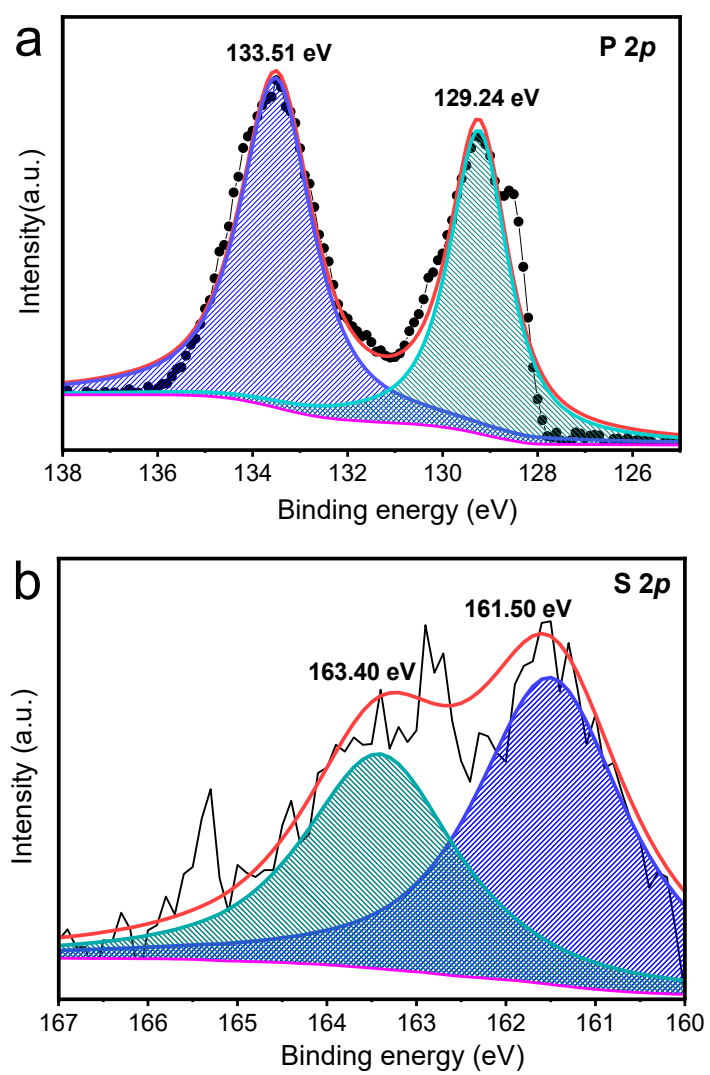
Supplementary Figure 9. (a) Solid-state ^{13}C -NMR of pure PCOS, (b) Schematic diagram of the molecular structure of PCOS.



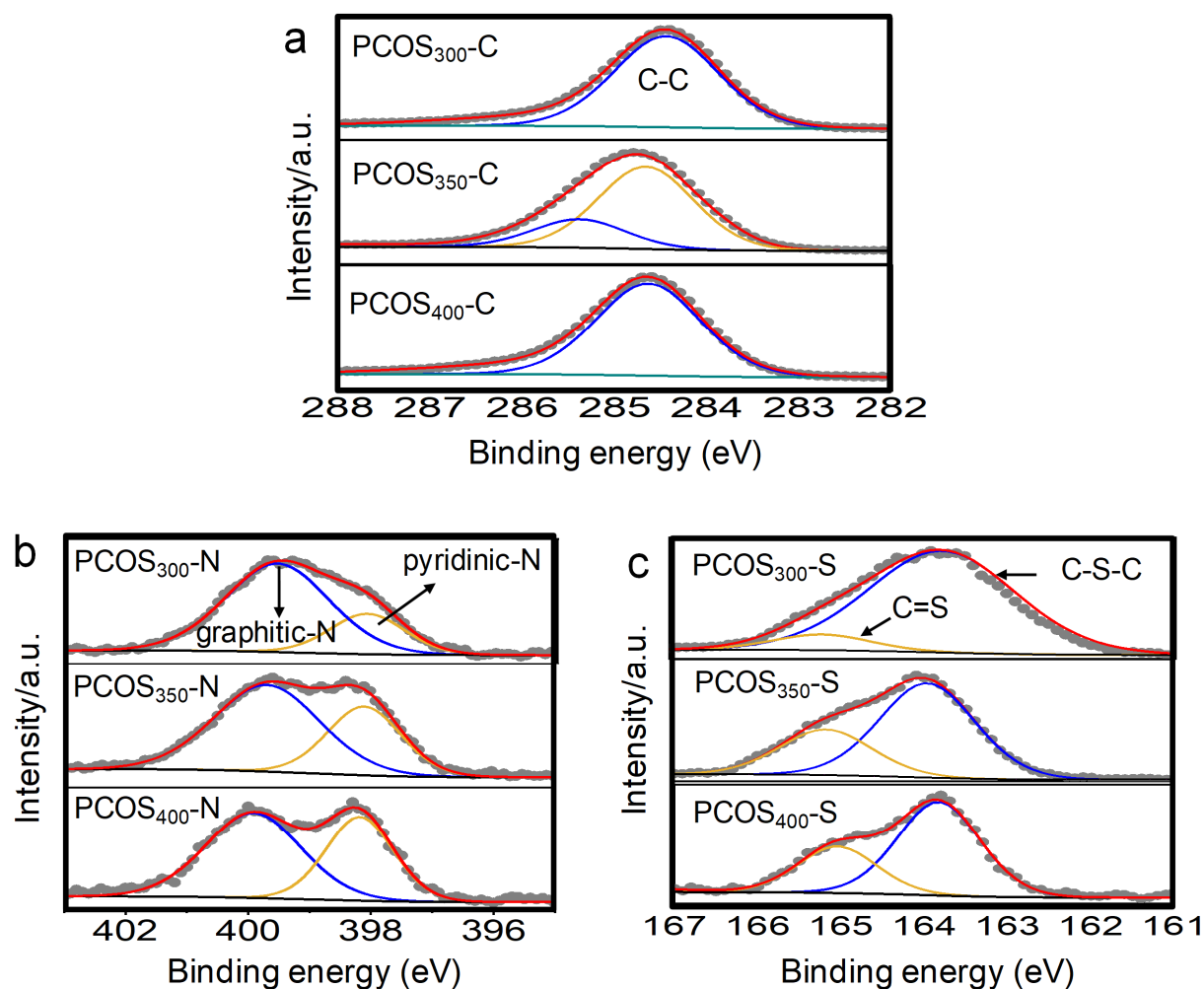
Supplementary Figure 10. XRD patterns of PCOS₃₀₀, PCOS₃₅₀, PCOS₄₀₀, PCOS₄₅₀, respectively.



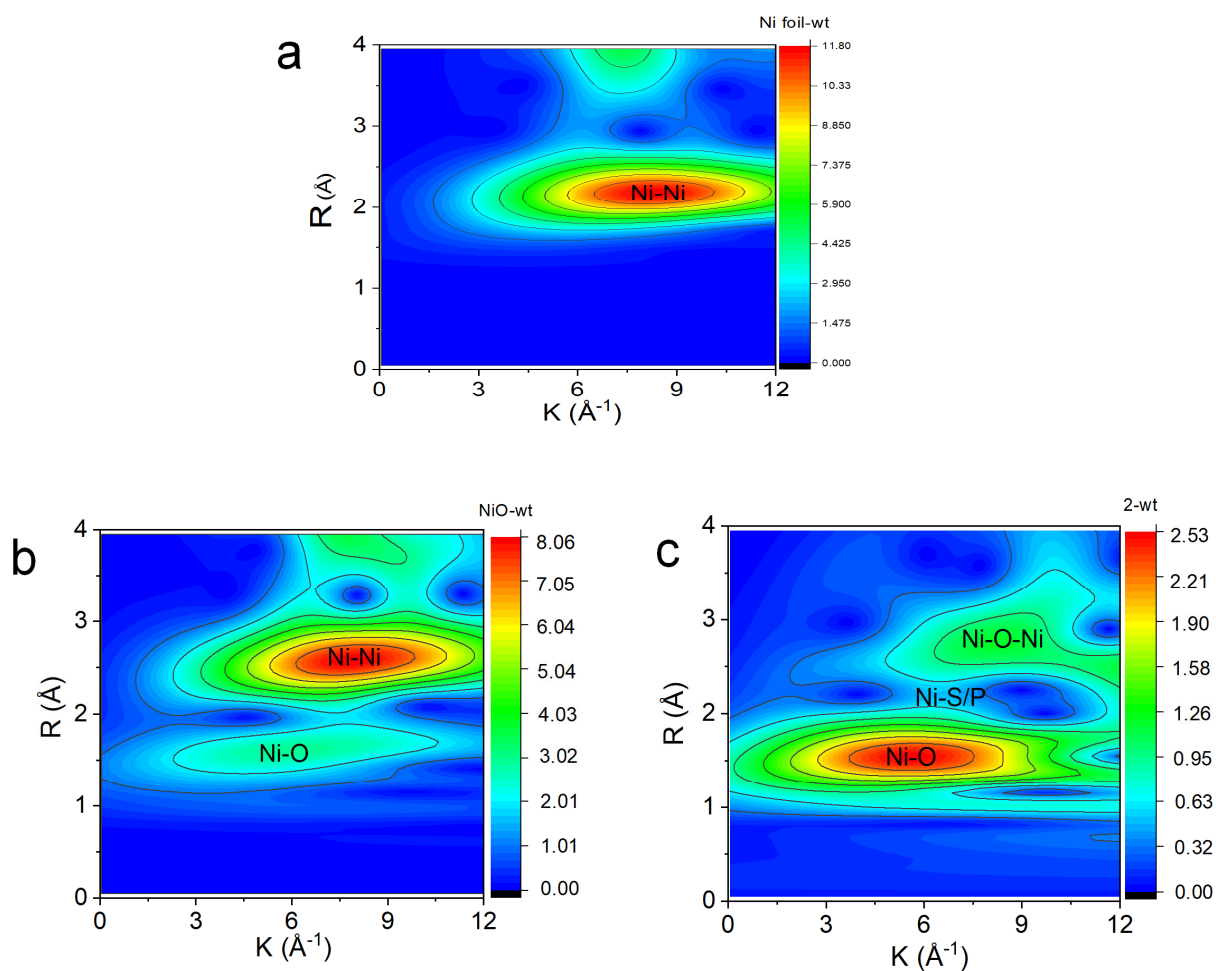
Supplementary Figure 11. FT-IR spectra of PCOS₃₀₀, PCOS₃₅₀, PCOS₄₀₀, PCOS₄₅₀, respectively.



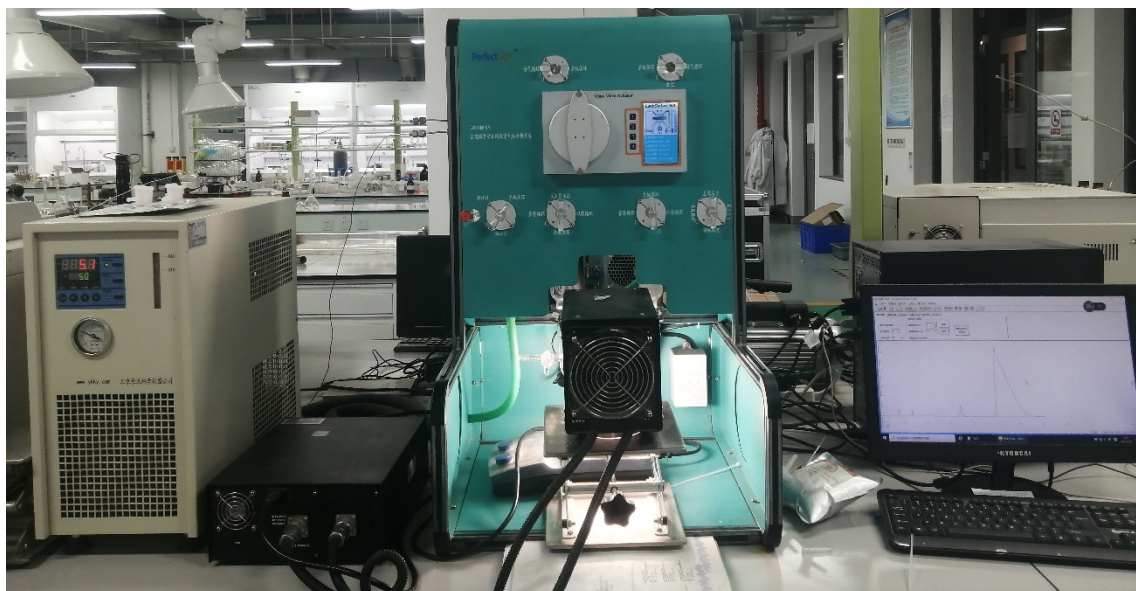
Supplementary Figure 12. XPS images of the Ni₂P/NiS@PCOS. (a) P 2p, (b) S 2p.



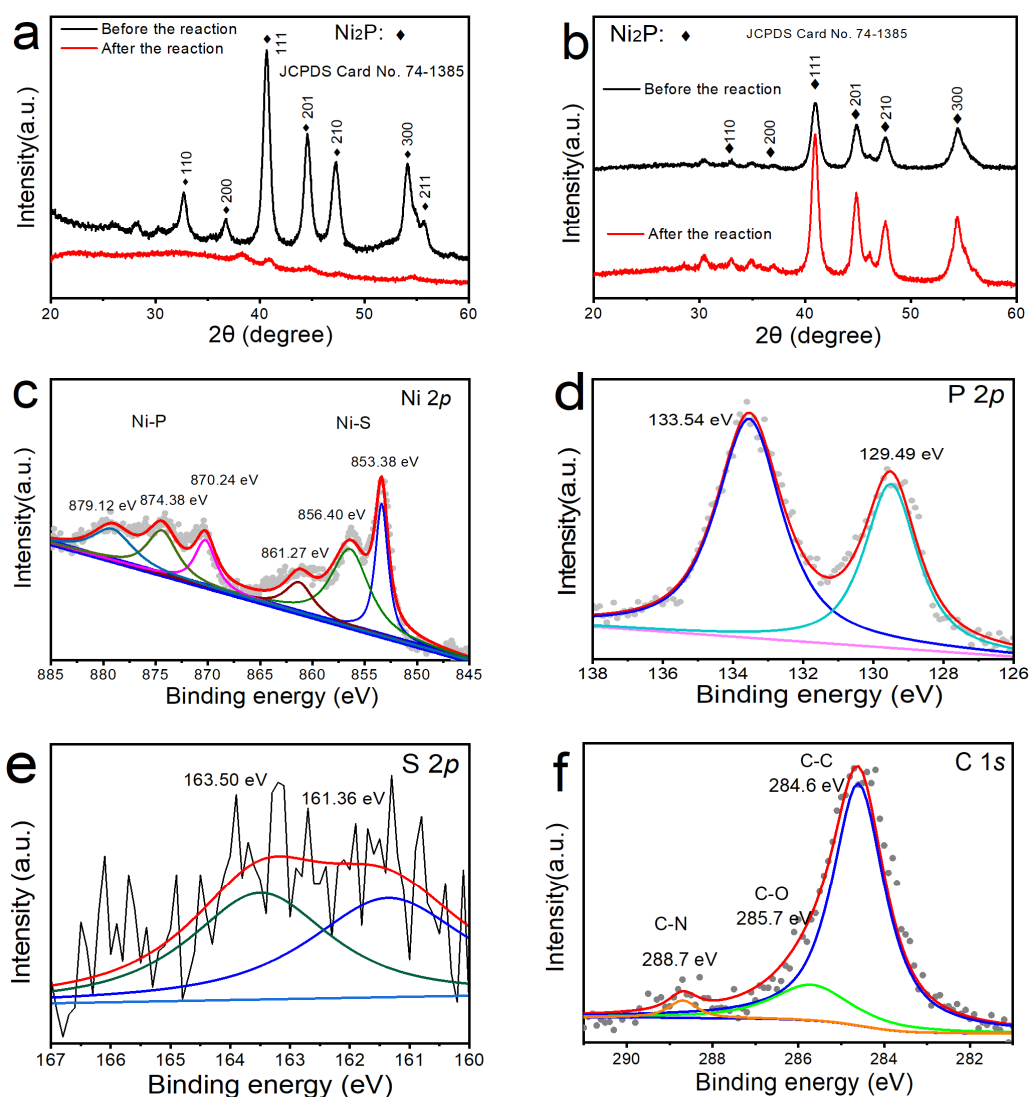
Supplementary Figure 13. XPS images of the polymeric carbon-oxygen semiconductor, which is the pure L-cysteine treated at 300 °C -400 °C for 3 h under nitrogen protection



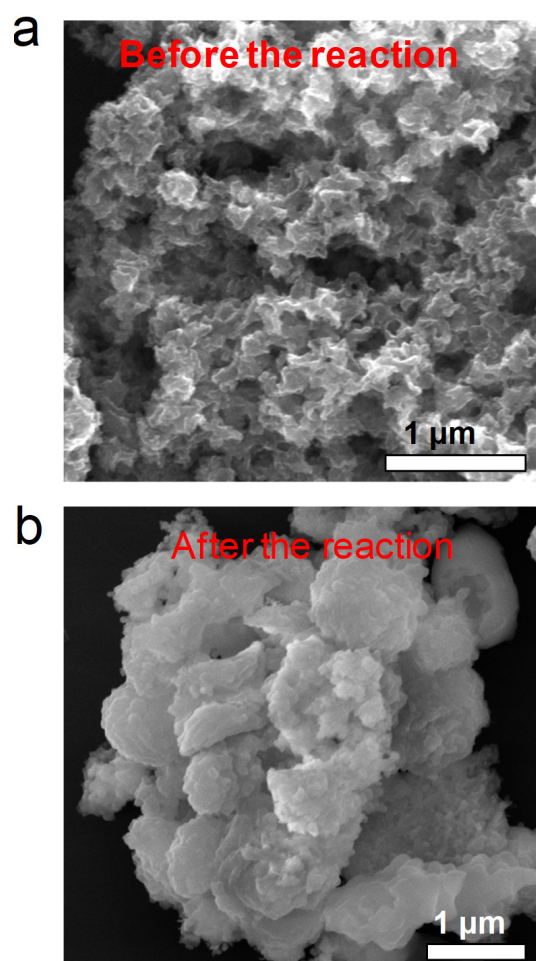
Supplementary Figure 14. Wavelet transform for the k^2 -weighted Ni K-edge EXAFS signal of (a) Ni standards, (b) NiO standards and (c) $\text{Ni}_2\text{P}/\text{NiS}$ sample in this work.



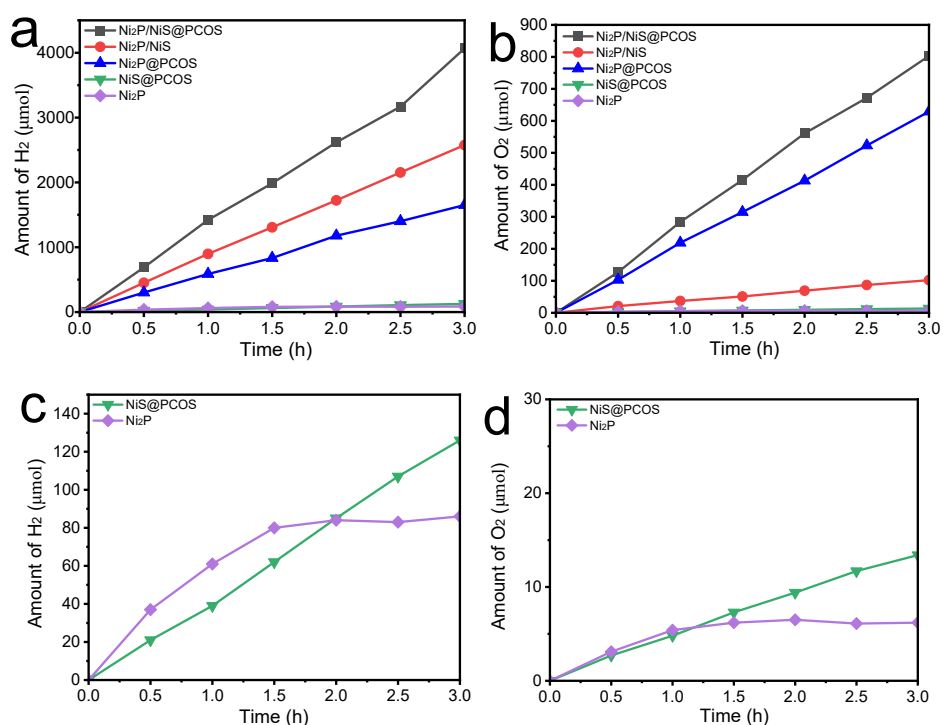
Supplementary Figure 15. Picture of photocatalytic overall water splitting test system.



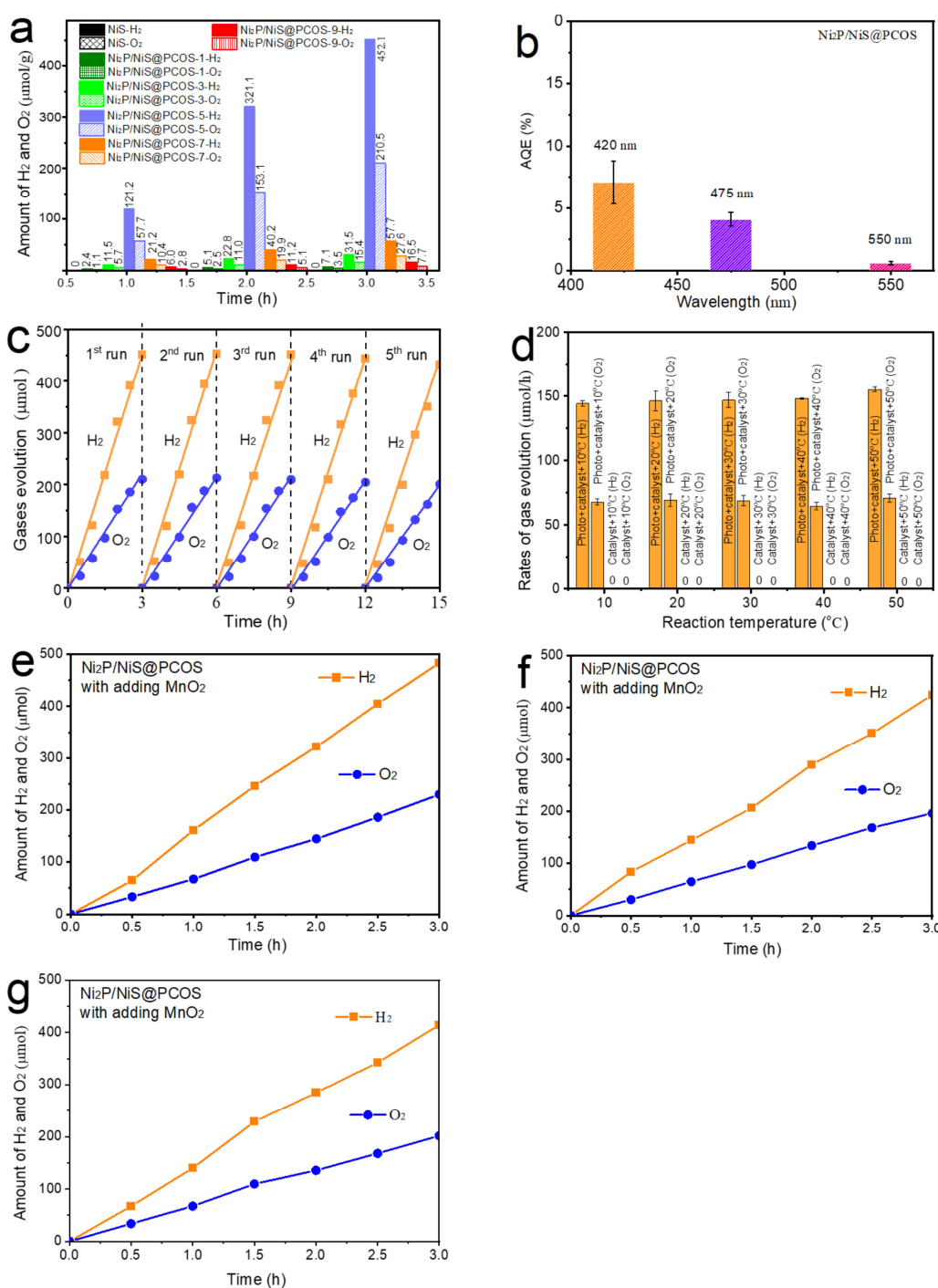
Supplementary Figure 16. (a) XRD patterns of the $\text{Ni}_2\text{P}/\text{NiS}@PCOS$ samples without adding MnO_2 (before the reaction and after 3h reaction). (b) XRD patterns of the $\text{Ni}_2\text{P}/\text{NiS}@PCOS$ samples with adding MnO_2 (before the reaction and after 3h reaction). (c-f) XPS spectra of the $\text{Ni}_2\text{P}/\text{NiS}@PCOS$ samples with adding MnO_2 after 3h reaction.



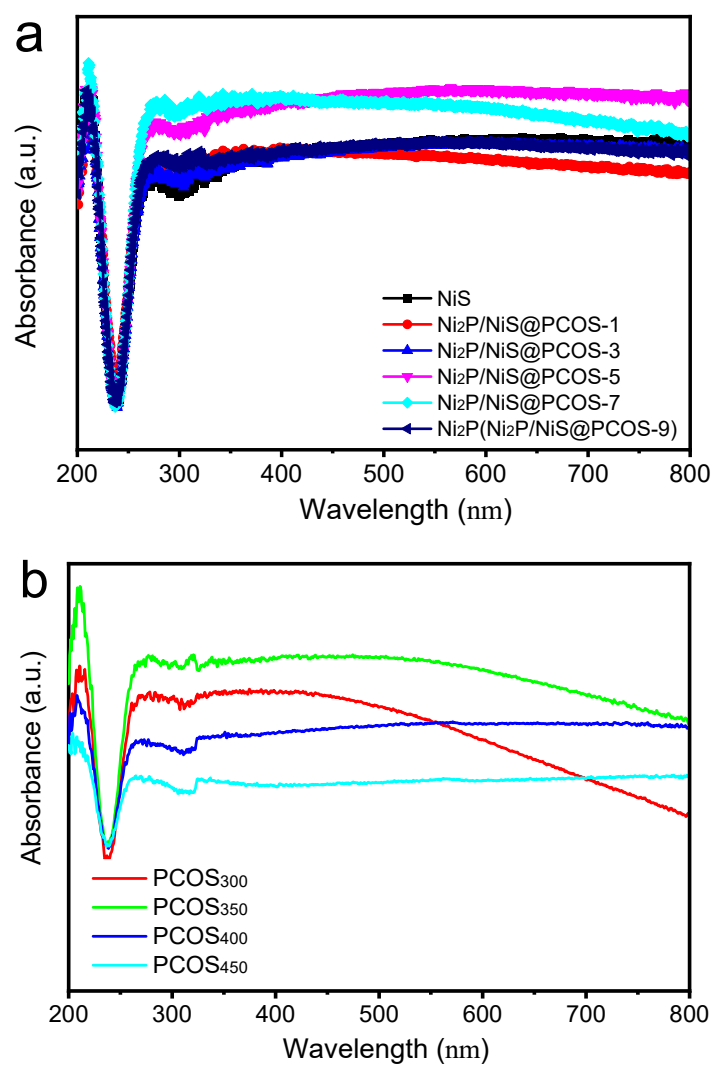
Supplementary Figure 17. The SEM of $\text{Ni}_2\text{P}/\text{NiS}@ \text{PCOS}$ sample without adding MnO_2 , (a) before the reaction, (b) after 3h reaction.



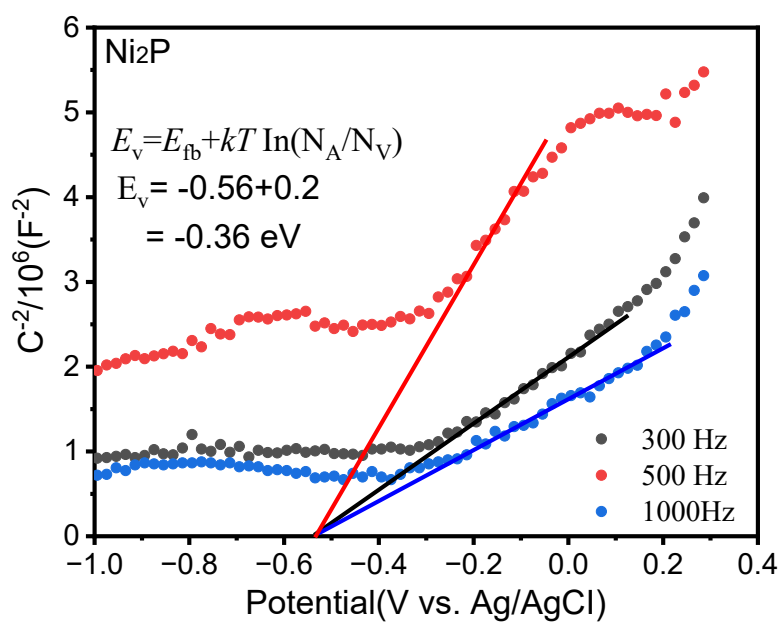
Supplementary Figure 18. (a,c) Time-dependent photocatalytic H₂ production of Ni₂P/NiS@PCOS, Ni₂P/NiS, Ni₂P/PCOS, NiS@PCOS and Ni₂P samples with lactic acid as holes sacrificial agents, (b,d) Time-dependent photocatalytic O₂ production of Ni₂P/NiS@PCOS, Ni₂P/NiS, Ni₂P/PCOS, NiS@PCOS and Ni₂P samples with AgNO₃ as electrons sacrificial agents. Experimental conditions: visible-light irradiation ($\lambda > 420$ nm), 100 mg photocatalyst, 10 mg MnO₂, 100 mL deionized water.



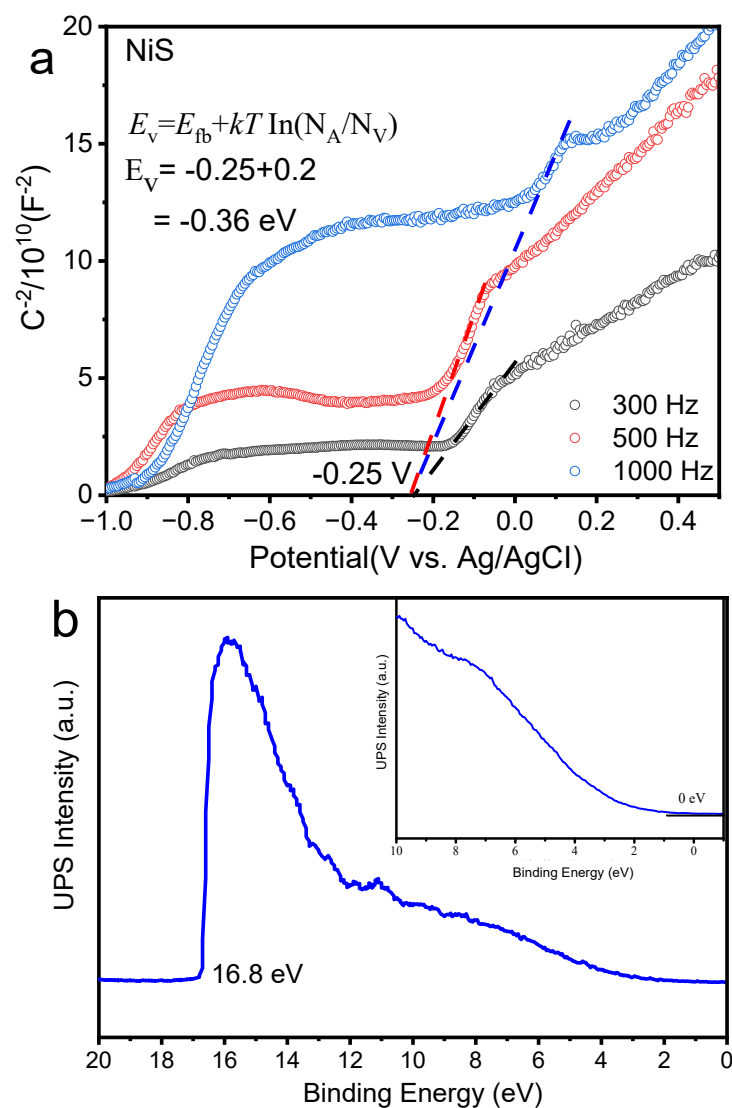
Supplementary Figure 19. (a) Time-dependent photocatalytic H₂ and O₂ production of Ni₂P/NiS@PCOS-x (0, 1, 3, 5, 7, 9) sample with MnO₂. (b) Wavelength-dependent AQE for photocatalytic H₂ evolution over Ni₂P/NiS@PCOS, (c) Stability tests of Ni₂P/NiS@PCOS for 5 runs, (d) Hydrogen/oxygen production over Ni₂P/NiS@PCOS under photothermocatalytic (10–50 °C) and thermo reaction (10–50 °C) conditions. (e-g) Three independent repeated experiments (tested by assistant professor He Li, PhD Baorong Xu and PhD Yuxi Ren. Experimental conditions: visible-light irradiation ($\lambda > 420$ nm). photocatalyst 100 mg, deionized water 100 mL, and 10 mg MnO₂ assisted by continuous stirring.) Error bars indicate standard deviation for three measurements.



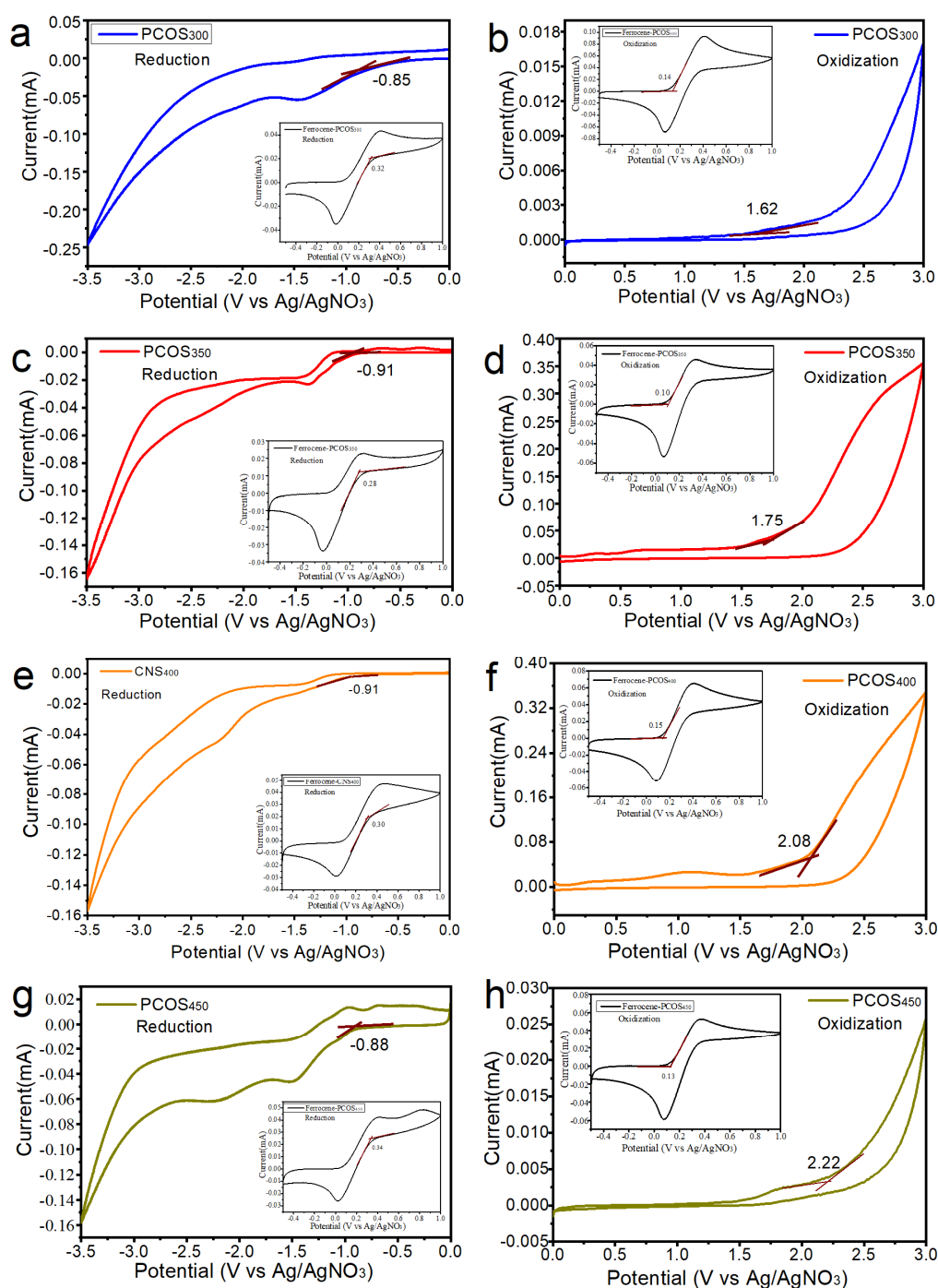
Supplementary Figure 20. UV–vis diffuse reflectance spectra (DRS) of Ni₂P, NiS, Ni₂P/NiS@PCOS and PCOS samples.



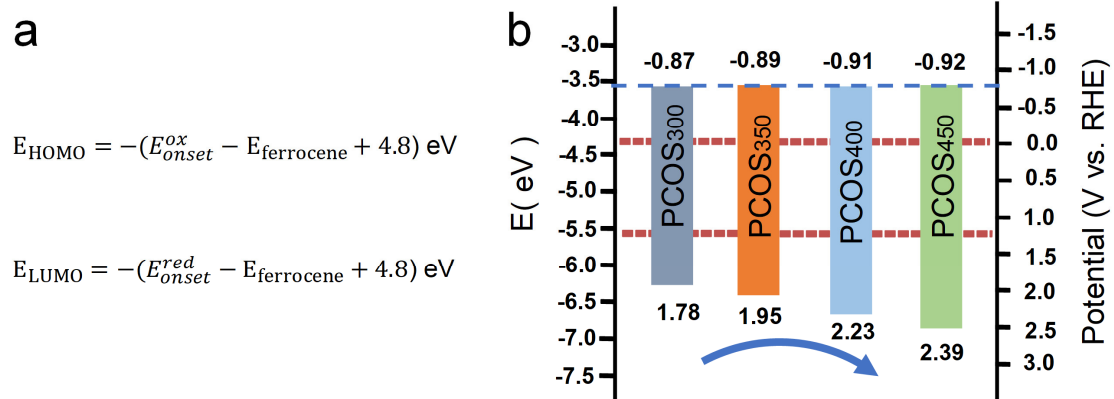
Supplementary Figure 21. Mott-Schottky plots of pure Ni₂P in 0.5 mol L⁻¹ Na₂SO₄ electrolytes (pH=7).



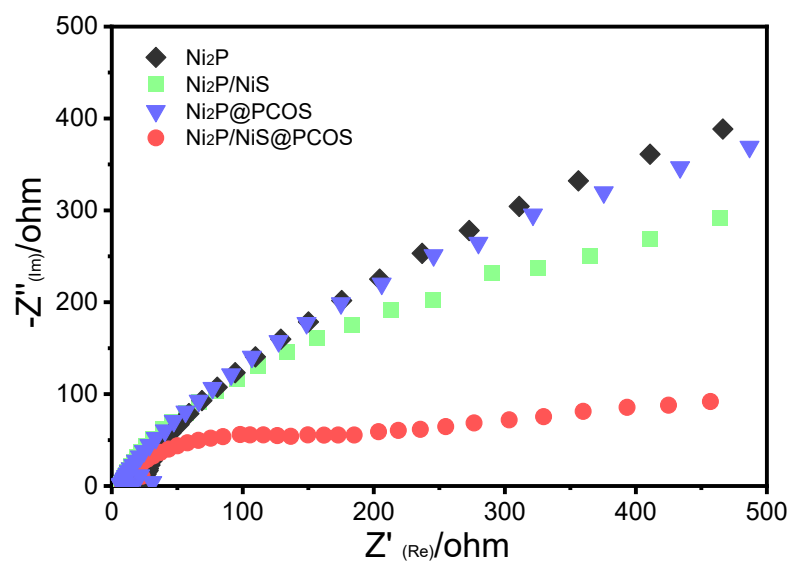
Supplementary Figure 22. Mott-Schottky plots of pure NiS in 0.5 mol L⁻¹ Na₂SO₄ electrolytes (pH=7); (b)UPS spectra of pure NiS.



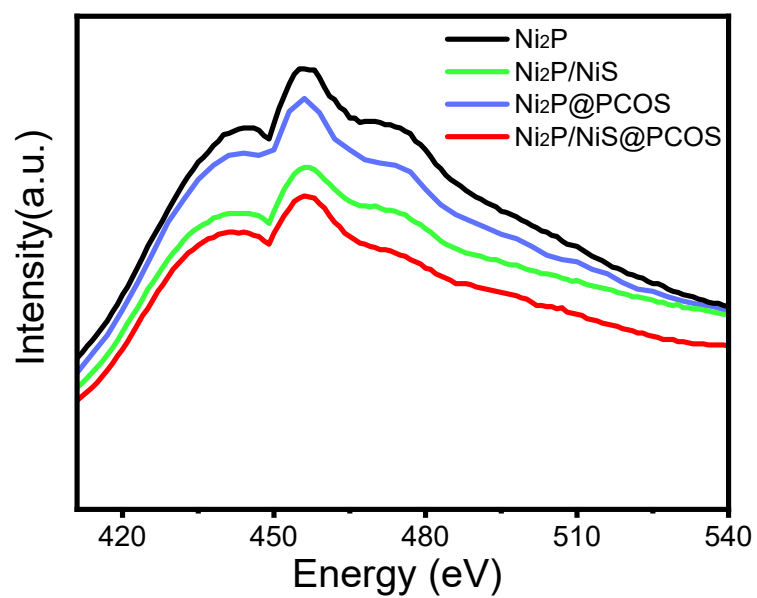
Supplementary Figure 23. Cyclic voltammograms of PCOS₃₀₀ (a-b), PCOS₃₅₀(c-d), PCOS₄₀₀(e-f), PCOS₄₅₀(g-h), respectively.



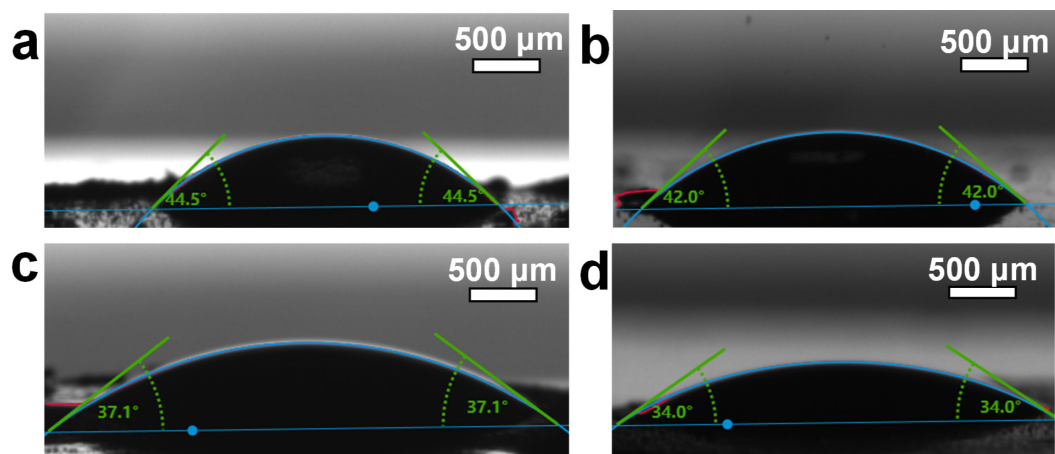
Supplementary Figure 24. (a-b) band structure diagram of PCOS₃₀₀, PCOS₃₅₀, PCOS₄₀₀, PCOS₄₅₀, respectively.



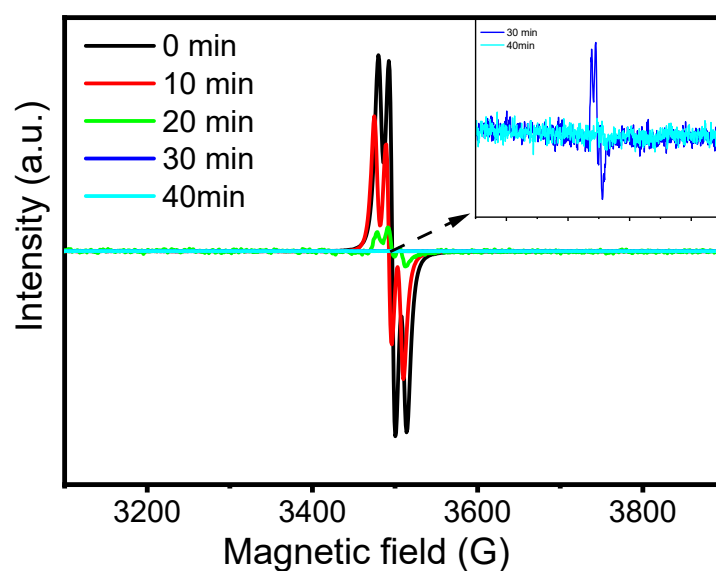
Supplementary Figure 25. Nyquist plots for the as-synthesized samples.



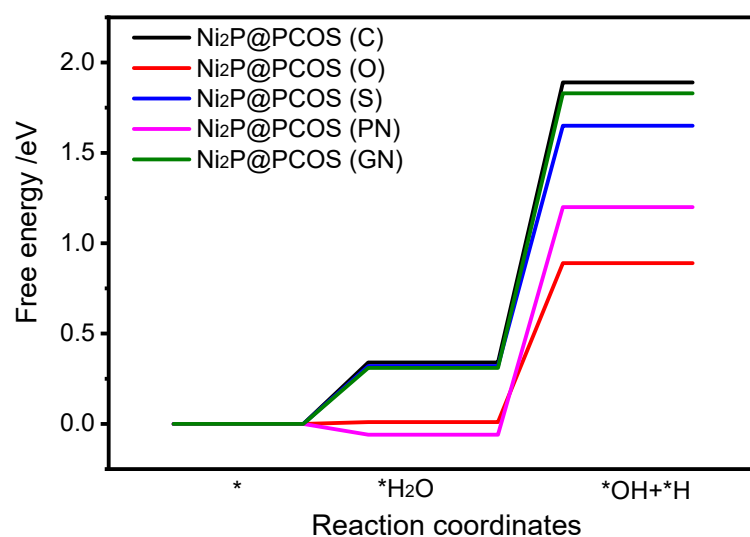
Supplementary Figure 26. PL spectra of various photocatalysts.



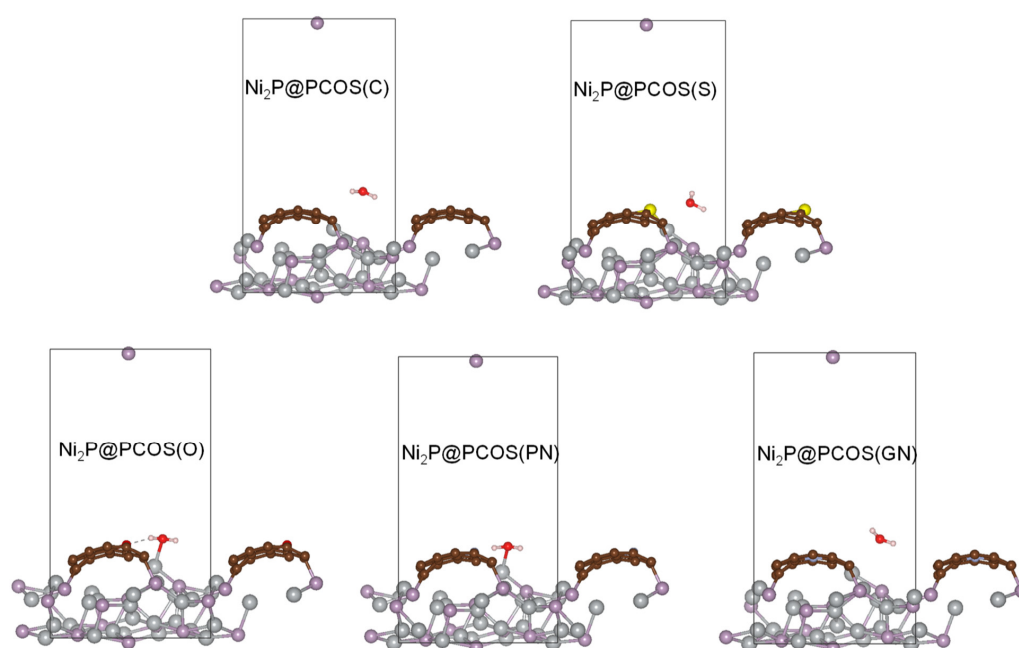
Supplementary Figure 27. Contact angle measurements of (a) Ni₂P, (b) Ni₂P/NiS, (c) Ni₂P@PCOS and (d) Ni₂P/NiS@PCOS.



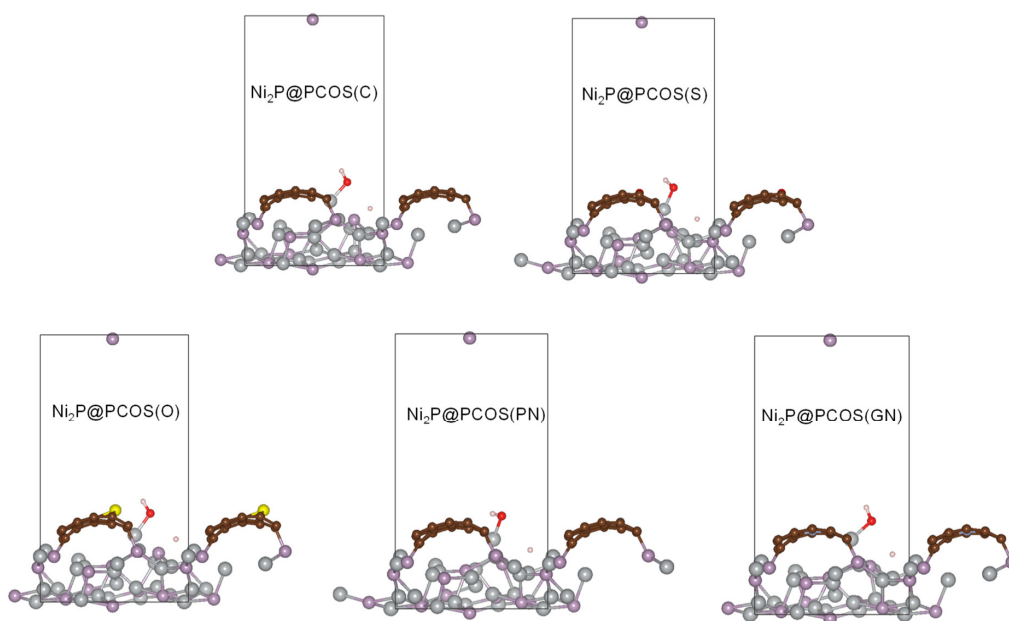
Supplementary Figure 28. ESR responses of the TEMPO-OH spin adduct in a Ni_2P water suspension under visible-light irradiation.



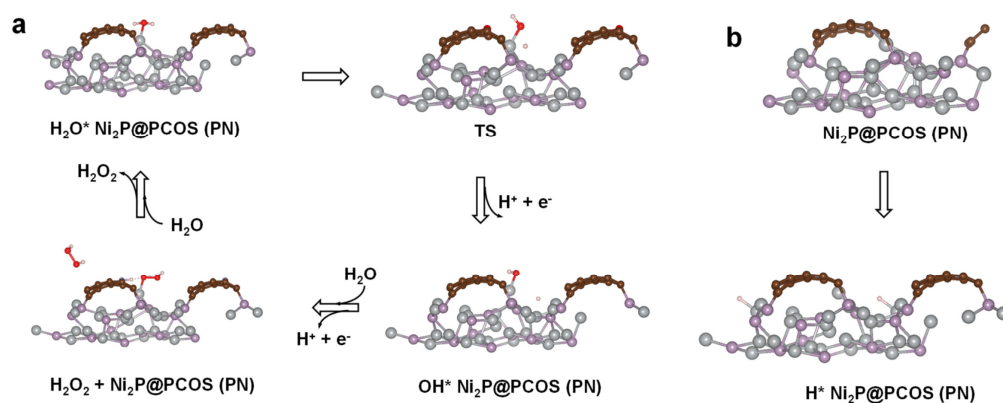
Supplementary Figure 29. Reaction energy for H₂O adsorption and dissociation on Ni₂P@PCOS structures.



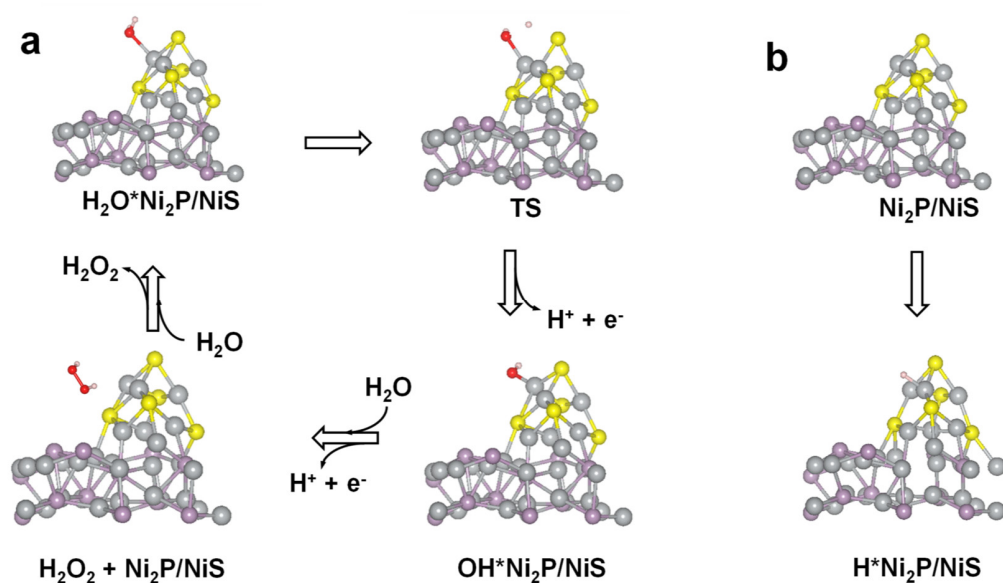
Supplementary Figure 30. Model diagram of adsorption water molecule on Ni₂P@PCOS surface. The yellow, red, pink, violet, brown and gray spheres represent S, O, H, P, C and Ni atoms, respectively.



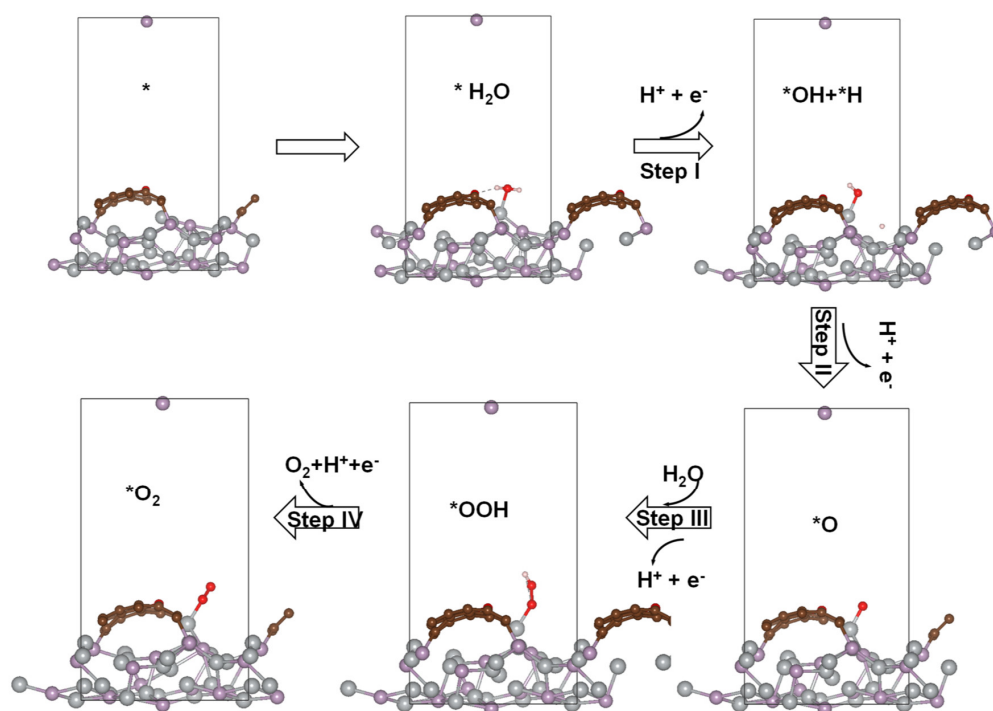
Supplementary Figure 31. Model diagram of dissociation water molecule on Ni₂P@PCOS surface. The yellow, red, pink, violet, brown and gray spheres represent S, O, H, P, C and Ni atoms, respectively.



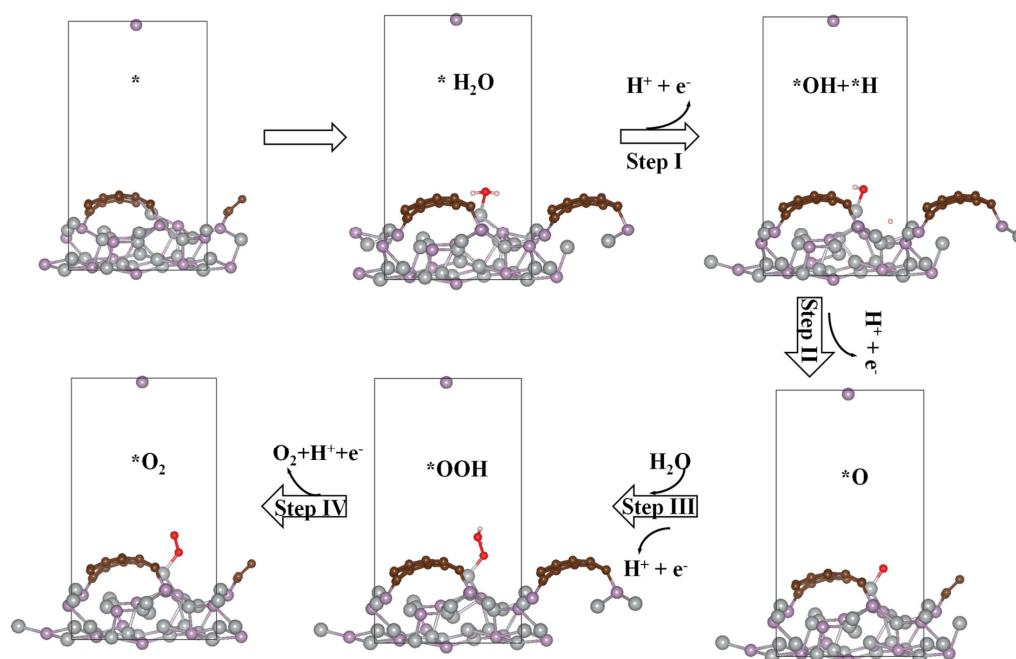
Supplementary Figure 32. (a) The optimized geometries of key reaction intermediates of 2e-mechanism of oxygen evolution reaction on $\text{Ni}_2\text{P@PCOS}$ (pyridine nitrogen sites), (b) Hydrogen adsorption model diagram on $\text{Ni}_2\text{P@PCOS}$ (pyridine nitrogen sites) surface. The yellow, red, pink, violet, brown and gray spheres represent S, O, H, P, C and Ni atoms, respectively.



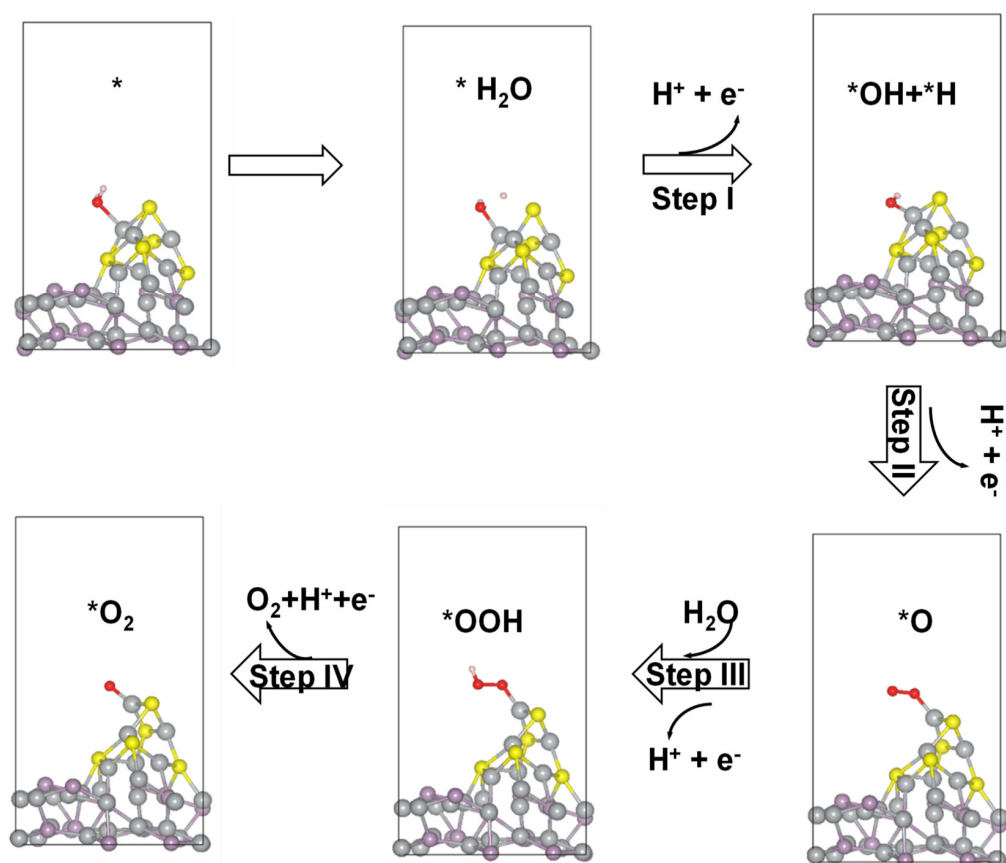
Supplementary Figure 33. (a) The optimized geometries of key reaction intermediates of 2e-mechanism of oxygen evolution reaction on $\text{Ni}_2\text{P}/\text{NiS}$, (b) Hydrogen adsorption model diagram on $\text{Ni}_2\text{P}/\text{NiS}$ surface. The yellow, pink, violet and gray spheres represent S, H, P and Ni atoms, respectively.



Supplementary Figure 34. The optimized geometries of key reaction intermediates of 4e⁻ mechanism of oxygen evolution reaction on Ni₂P@PCOS (O atom sites) surface. The yellow, red, pink, violet, brown and gray spheres represent S, O, H, P, C and Ni atoms, respectively.



Supplementary Figure 35. The optimized geometries of key reaction intermediates of 4e⁻ mechanism of oxygen evolution reaction on Ni₂P@PCOS (pyridine nitrogen sites) surface. The yellow, red, pink, violet, brown and gray spheres represent S, O, H, P, C and Ni atoms, respectively.



Supplementary Figure 36. The optimized geometries of key reaction intermediates of 4 e^- mechanism of oxygen evolution reaction on $\text{Ni}_2\text{P}@ \text{NiS}$ surface. The yellow, red, pink, violet and gray spheres represent S, O, H, P and Ni atoms, respectively.

Supplementary Table 1 BET surface area, pore volume and pore size of Ni₂P/NiS@PCOS sample.

Samples	BET surface area (m ² g ⁻¹)	Pore volume (cm ³ g ⁻¹)	Pore width (nm)
Ni ₂ P/NiS@PCOS	29	0.2	1.33

Supplementary Table 2 The content of NiS, NiP and PCOS in the samples of Ni₂P/NiS@PCOS-x (0,1,3,5,7,9) were figured out by EDX.

samples	NiS	Ni ₂ P	PCOS
NiS@PCOS	61.66%	0%	38.34%
Ni ₂ P/NiS@PCOS-1	35.53%	29.53%	34.94%
Ni ₂ P/NiS@PCOS-3	23.50%	44.88%	31.62%
Ni ₂ P/NiS@PCOS-5	9.71%	52.88%	37.41%
Ni ₂ P/NiS@PCOS-7	2.66%	66.69%	30.65%
Ni ₂ P/NiS@PCOS-9	1.32%	71.06%	27.62%

Supplementary Table 3 Atomic percentage of polymeric carbon-oxygen semiconductor by XPS testing.

samples	C element	N element	S element	O element
PCOS ₃₀₀	86.04 %	7.49 %	2.97 %	3.50 %
PCOS ₃₅₀	82.39 %	6.44 %	2.87 %	8.30 %
PCOS ₄₀₀	85.46 %	5.24 %	1.98 %	7.32 %

Supplementary Table 4 Atomic percentage of polymeric carbon-oxygen semiconductor by EDX testing.

samples	C element	N element	S element	O element
PCOS ₃₀₀	67.7%	21 %	0.57%	10.6 %
PCOS ₃₅₀	63.4 %	20.9 %	0.48 %	15.1 %
PCOS ₄₀₀	67.4%	21 %	0.22 %	11.2 %

Supplementary Table 5. EXAFS fitting parameters at the Ni K-edge for various samples ($S_0^2=0.75$)

^a*N*: coordination numbers; ^b*R*: bond distance; ^c σ^2 : Debye-Waller factors; ^d ΔE_0 : the inner potential correction. *R* factor: goodness of fit.

samples	shell	CN	R(Å)	σ^2	ΔE_0	R factor
Ni foil	Ni-Ni	12	2.48±0.01	0.0062	6.4±0.3	0.0012
Ni ₂ P/NiS/PCOS	Ni-P/S	4.7±0.2	2.23±0.01	0.0075	-5.1±1.3	0.0044
	Ni-Ni	2.9±0.2	2.61±0.02	0.0075		
	Ni-S-Ni	1.7±0.3	2.96±0.01	0.0075		
Ni ₂ P/NiS	Ni-O	5.1±0.4	2.01±0.01	0.0054	-7.7±1.9	0.0084
	Ni-S/P	0.7±0.2	2.33±0.03	0.0029		
	Ni-O-Ni	1.3±0.2	3.04±0.02	0.0026		

Supplementary Table 6 Comparison of the overall water splitting activity of Ni₂P/NiS@PCOS and other typical materials.

Year	Catalyst	Light source	HER μmol h ⁻¹ g ⁻¹	OER μmol h ⁻¹ g ⁻¹	AQE	Author	Journal Title	Ref.
1980	NiO-SrTiO ₃	450 W	0.196	0.098	—	Domen K	Journal of the Chemical Society-Chemical Communications	1
1988	NiO(1wt%)-K ₄ Nb ₆ O ₁₇	450 W	66	38	—	Kudo A	Journal of catalysis	2
2001	Pt-loaded TiO ₂ -anatase/TiO ₂ -rutile	>300nm 400 W	180	90	—	Abe R	Chemical physics letters	3
2007	(Ga _{1-x} Zn _x)(N _{1-x} O _x)RuO ₂	>400nm 450 W	200	100	30%, 600 nm	Maeda K	The Journal of Physical Chemistry C	4
2013	IrO ₂ /Cr ₂ O ₃ /RuO _x /ZrO ₂ /TaON	>400nm 450 W	14	7	0.1% 420nm	Maeda K	Chemistry—A European Journal	5
2015	CDots-C ₃ N ₄	>420nm 300 W	8.4	4.1	16%, 420nm	Liu J	Science	6
2015	In _{0.26} Ga _{0.74} N: Mg	>400nm 300 W	1000	500	12.3% (400–475 nm) 200 °C	Kibria M G,	Nature communications	7
2016	Cr ₂ O ₃ / Ba ₅ Ta ₄ O ₁₅	UV 500 W	930	456	—	Julia Soldat	ChemCatChem	8
2016	Al-SrTiO ₃	> 300 nm	5500	2800	30%, 360nm	Ham Y	Journal of Materials Chemistry A	9
2017	PtOx/WN	>420nm 300 W	0.90	0.48	—	Yu Lei	Angewandte Chemie-International Edition	10
2018	aza-CMP/C ₂ N	420	5	2.5	4.3% 600 nm	Wang L	Angewandte Chemie-International Edition	11
2018	Rh/Cr ₂ O ₃ -T a ₃ N ₅ /KTaO ₃	>420nm 300 W	36.7	15.3	0.22% 420nm	Wang Z	Nature Catalysis	12
2018	RuDTC:Pt@CdS	460 ± 50 nm	170	71	0.27%	Wolff C M	Nature Energy	13
2019	Fe ₂ O ₃ /RGO/PCN	>420nm 300 W	1090	530	—	Pan Z	Angewandte Chemie-International Edition	14
2020	Pt@NH ₂ -UiO-66@MnOx	>400nm 300 W	19.6	10.1	—	Zhang J	Advanced Materials	15
2021	Pt/TiO ₂ /CdS-ZCGSe/Au/BiVO ₄ :Mo	>420nm 300 W	200	100	1.5% 420nm	Chen S,	Journal of the American Chemical Society	16
2021	BiOBr/C	>420nm 300 W	240	110	1.456% 420nm	Xiaoqi Zheng	ACS Nano	17
2021	BDCNN ₃₅₀ /BDCNN ₄₂₅	>300nm 300 W	823.5	410.5	10.44% 420nm	Zhao D	Nature Energy	18

Supplementary Table 7. The calculated parameters about the corresponding energy-level structure of the different samples.

Samples	$E_{\text{cut off}}/\text{eV}$	W_s/eV	E/eV	E_{VB}/eV	E_g/eV	E_{CB}/eV
Ni ₂ P	16.70	-4.52	-1.90	-6.42	2.34	-4.08
NiS	16.80	-4.42	-0.00	-4.42	0.03	-4.39

Supplementary Table 8 Comparison of the photocatalytic hydrogen production activity of pure Ni₂P or CoxP in the half water splitting.

	Transition metal phosphide	Band gap	photocatalytic hydrogen production activity	References
1	Ni ₂ P	1.38 eV	untested	[19] Chemical Physics Letters 730 (2019) 478–484
2	Ni ₂ P	1.0 eV	16 μmol/10mg/h	[20] Catalysis Letters (2019) 149:3012–3026
3	Ni ₂ P	untested	12 mmol/g/h	[21] Chem. Commun., 2018, 54, 6188
4	Ni ₂ P	untested	2.19 μmol/10mg/h	[22] Journal of Colloid and Interface Science 537 (2019) 629–639
5	Ni ₂ P	untested	258.6 μmol/g/h	[23] <i>Chem. Eng. J.</i> 2021, 409, 128184
6	Ni ₂ P	untested	1250 μmol/g/h	[24] <i>Angew. Chem. Int. Ed.</i> 2018,57, 13082-13085
7	Co ₂ P	untested	136 μmol/8mg/h	[25] J. Phys. Chem. C 2016, 120, 6409–6415
8	Co ₂ P	0.74 eV	0.13 mmol/g/h	[26] ACS Appl. Mater. Interfaces 2019, 11, 22297-22306
9	CoP	1.71 eV	untested	[27] Adv. Energy Mater. 2017, 7, 1602355

Supplementary Table 9. The frequency of H₂O molecule at transfer states for dissociation on different sites.

Frequency	$f_{\text{TS}_{\text{Ni}_2\text{P@PCOS (O)}}}/\text{meV}$	$f_{\text{TS}_{\text{Ni}_2\text{P@PCOS (PN)}}}/\text{meV}$	$f_{\text{TS}_{\text{Ni}_2\text{P@NiS}}}/\text{meV}$
1 f	443.16	459.1634	466.1172
2 f	236.29	237.9481	118.3730
3 f	110.07	99.31787	84.3548
4 f	70.94	66.33454	72.2608
5 f	53.97	35.86591	58.0672
6 f	47.42	24.02567	48.0154
7 f	26.73	16.30477	15.0993
8 f	19.52	8.115175	3.4150
9 f/i	86.49	63.89422	179.3769

Supplementary Table 10. Adsorption energy of H₂O on Ni₂P@PCOS sites.

Site/doping element	Adsorption energy/eV
C	0.12
Ni	-0.08
Ni/O	-0.54
Ni/S	-0.11
Ni/pyridinic-N	-0.6
Ni/graphitic-N	-0.05

Supplementary Table 11. Gibbs free-energy value for the two steps of H₂O₂ evolution on Ni₂P@PCOS (PN)

Intermediate product	$\Delta E_{\text{ads}} / \text{eV}$	$\Delta \text{ZPE} / \text{eV}$	$T\Delta S / \text{eV}$	$\Delta G / \text{eV}$	G / eV
*	—	—	—	—	0
*H ₂ O	-0.619	0.045	-0.509	-0.065	-0.065
*TS	1.847	-0.158	0.063	1.626	1.560
*OH	1.377	-0.097	0.020	1.260	1.195
*H ₂ O ₂	-1.004	0.213	0.082	-0.873	0.322
*	0.576	0.035	0.272	0.339	0.661

Supplementary Table 12. Gibbs free-energy value for the four steps of O₂ evolution on Ni₂P@PCOS (PN)

Intermediate product	ΔE_{ads} /eV	ΔZPE /eV	$T\Delta S$ /eV	ΔG /eV	G /eV
*	—	—	—	—	0
*H ₂ O	-0.619	0.045	-0.509	-0.065	-0.065
*TS	1.847	-0.158	0.063	1.626	1.560
*OH	1.377	-0.097	0.020	1.260	1.195
*O	1.116	0.421	-0.006	1.543	2.738
*OOH	-1.628	0.374	0.114	-1.368	1.370
*OO	0.798	-0.299	-0.079	0.578	1.948
*	0.231	-0.023	0.497	-0.289	1.659

Supplementary Table 13. Gibbs free-energy value for the two steps of H₂O₂ evolution on Ni₂P@PCOS (O)

Intermediate product	ΔE_{ads} /eV	ΔZPE /eV	$T\Delta S$ /eV	ΔG /eV	G /eV
*	—	—	—	—	0
*H ₂ O	-0.558	0.090	-0.478	0.010	0.010
*TS	1.355	-0.172	-0.027	1.210	1.220
*OH	1.07	-0.162	0.029	0.877	0.887
*H ₂ O ₂	-0.407	0.193	0.071	-0.285	0.603
*	0.225	0.004	0.171	0.058	0.661

Supplementary Table 14. Gibbs free-energy value for the four steps of O₂ evolution on Ni₂P@PCOS (O)

Intermediate product	ΔE_{ads} /eV	ΔZPE /eV	$T\Delta S$ /eV	ΔG /eV	G /eV
*	—	—	—	—	0
*H ₂ O	-0.558	0.090	-0.478	0.010	0.010
*TS	1.355	-0.172	-0.027	1.210	1.220
*OH	1.07	-0.162	0.029	0.877	0.887
*O	1.876	-0.460	-0.050	1.467	2.354
*OOH	-1.602	0.371	0.062	-1.293	1.061
*OO	0.982	-0.294	0.038	0.650	1.712
*	0.411	-0.026	0.438	-0.053	1.658

Supplementary Table 15. Gibbs free-energy value for the two steps of H₂O₂ evolution on Ni₂P/NiS

Intermediate product	ΔE_{ads} /eV	ΔZPE /eV	$T\Delta S$ /eV	ΔG /eV	G /eV
*	—	—	—	—	0
*H ₂ O	-0.563	0.062	-0.454	-0.047	-0.047
*TS	2.117	-0.125	-0.068	2.060	2.013
*OH	-1.237	-0.190	0.076	-1.503	0.510
*H ₂ O ₂	-0.667	0.104	-0.544	-0.019	0.491
*	0.680	0.037	0.547	0.170	0.661

Supplementary Table 16. Gibbs free-energy value for the four steps of O₂ evolution on Ni₂P/NiS

Intermediate product	ΔE_{ads} /eV	ΔZPE /eV	$T\Delta S$ /eV	ΔG /eV	G /eV
*	—	—	—	—	0
*H ₂ O	-0.563	0.062	-0.454	-0.047	-0.047
*TS	2.117	-0.125	-0.068	2.060	2.013
*OH	-1.237	-0.190	0.076	-1.503	0.510
*O	1.469	-0.271	-0.069	1.267	1.777
*OOH	-1.901	0.125	-0.482	-1.294	0.483
*OO	0.919	-0.284	-0.058	0.693	1.176
*	1.037	-0.035	0.525	0.477	1.653

Supplementary Table 17. The atoms coordinate of Ni₂P/NiS model.

Atom	X coordinate/Å	Y coordinate/Å	Z coordinate/Å
Ni	2.6712	6.8513	0.7803
Ni	7.0032	1.7750	0.8200
Ni	0.2312	6.6445	3.1643
Ni	4.4987	1.6518	3.2679
Ni	2.5458	3.3068	1.0055
Ni	7.5734	8.4878	0.0830
Ni	8.8402	3.1403	3.2793
Ni	5.8460	8.3212	2.6307
Ni	3.8481	0.0321	0.0075
Ni	8.3138	5.1103	0.0898
Ni	1.1035	10.1393	2.0491
Ni	5.6371	4.9664	2.1959
Ni	0.0298	3.0862	0.8890
Ni	4.4979	8.2919	0.8668
Ni	1.8391	8.0759	2.6906
Ni	6.3548	2.8626	2.7609
Ni	0.2645	7.0228	0.8523
Ni	4.5219	1.9597	0.9476
Ni	1.7903	2.0398	2.7592
Ni	7.0583	6.8103	3.6700
Ni	3.9957	4.9780	0.7607
Ni	8.6021	0.1908	0.3688
Ni	1.4487	4.6020	2.6772
Ni	6.0548	0.3011	2.5205
Ni	2.0628	1.2585	0.4175
Ni	6.6343	6.6773	1.0319
Ni	4.0015	6.6434	2.5860
Ni	8.3165	1.1940	2.5944
Ni	2.0479	8.9315	0.3520
Ni	6.5144	3.9133	0.4513
P	3.7790	3.6813	2.9859
P	8.1384	8.6870	2.3571
P	1.5564	5.0596	0.1922
P	6.0673	10.1206	0.0577
P	3.4145	10.1338	2.4402
P	7.9436	4.9343	2.3949
P	4.9749	7.7461	4.5818
P	4.5076	5.9093	6.4366
P	2.8512	7.9325	6.1976
P	2.1678	5.4163	6.6963
P	5.3056	5.1230	4.2952
P	2.0248	6.2535	4.3675
S	0.8754	6.9918	6.1361
S	3.1577	8.8201	4.3008
S	3.2923	6.7232	7.9752
S	3.3246	4.5299	5.0460
S	6.4197	6.4315	5.6983

Supplementary Table 18. The atoms coordinate of Ni₂P@PCOS model.

Atom	X coordinate/Å	Y coordinate/Å	Z coordinate/Å
Ni	8.2803	4.9695	18.0444
Ni	3.7175	9.9689	0.0236
Ni	6.8432	1.7427	0.1983
Ni	7.4698	8.4305	0.4163
Ni	2.4084	6.7095	0.7554
Ni	3.1974	3.2740	0.8117
Ni	0.7048	3.3078	2.0688
Ni	1.1882	10.0117	2.3774
Ni	6.0022	4.9981	2.3931
Ni	5.9134	0.9607	3.1010
Ni	4.4953	8.0607	3.3194
Ni	0.5120	6.6761	3.3065
Ni	5.9245	9.8818	0.0121
Ni	8.7910	2.8059	0.1583
Ni	1.5432	4.6753	0.2441
Ni	0.1642	6.8878	0.2871
Ni	4.6418	1.7340	0.3437
Ni	6.4875	3.9696	0.4385
Ni	1.8544	8.9157	0.4561
Ni	6.5016	6.5301	0.8149
Ni	8.3698	0.2353	0.8549
Ni	2.1254	1.2208	0.8561
Ni	4.2731	8.0285	1.0459
Ni	4.1010	5.2498	1.1752
P	8.3312	5.0992	2.0732
P	7.4357	2.4903	2.2527
P	8.6775	8.3590	2.2874
P	5.0060	3.0463	2.3196
P	6.1673	9.1028	2.3695
P	2.0628	5.2040	2.7623
P	2.3294	7.8016	2.7501
P	2.6216	2.6054	2.8612
P	3.3644	-0.0543	2.9582
P	4.1400	6.0981	4.3004
P	6.7338	6.8678	3.4169
P	8.1525	0.3782	3.4867
C	3.4361	1.4612	4.3067
C	4.8579	1.6216	4.4362
C	0.1423	1.4243	4.8509
C	7.7740	1.4880	4.9678
C	2.5452	2.4662	4.9668
C	0.1264	6.0186	4.8904
C	7.6772	6.0610	4.9459
C	1.0603	2.4615	5.1665
C	5.5090	2.5904	5.2741

C	2.4879	4.9563	4.9962
C	6.9527	2.5606	5.4028
C	1.0411	4.9291	5.2512
C	3.2527	3.7388	5.2117
C	0.3052	3.6997	5.4886
C	6.9177	4.9676	5.4621
C	5.4626	4.9946	5.3614
C	4.7150	3.7779	5.4287
C	7.7427	3.7466	5.5983

Supplementary Table 19. Basic parameters of AQE test

wavelength (m)	activity (mol/h)	Avogadro number (mol ⁻¹)	light intensity (w/cm ²)	Area (cm ²)	Reaction Time (s)	Planck's constant (J s)	Speed of light (m/s)
λ	V	N_A	I	A	t	h	c
425*10 ⁻⁹	34.7*10 ⁻⁶	6.02*10 ²³	1.02 ⁻³	78	3600	6.62*10 ⁻³⁴	3*10 ⁸
475*10 ⁻⁹	55.7*10 ⁻⁶	6.02*10 ²³	2.38 ⁻³	78	3600	6.62*10 ⁻³⁴	3*10 ⁸
550*10 ⁻⁹	7.1*10 ⁻⁶	6.02*10 ²³	1.8410 ⁻³	78	3600	6.62*10 ⁻³⁴	3*10 ⁸

Supplementary Table 20. Basic parameters of STH test

Rate of hydrogen production	The Gibbs free energy	Optical power density	Area of illumination
R_{H_2}	ΔG_r	P_{sun}	S
$3.84 \times 10^{-5} \text{ mmol s}^{-1}$	$237000 \text{ J mol}^{-1}$	100 mW cm^{-2}	10 cm^2

Supplementary note 1: experimental details for reproducibility

(1) Preparation of the NiS@L-cysteine nanosheets: The NiS@L-cysteine nanosheets were prepared by a one-step hydrothermal process. In this procedure, 2 mmol NiCl₂ and 2 mmol L-Cysteine are mixed in 25 mL deionized water, and continuously stirring 6 h (450 r/min). Then, 10 mL solution with a certain concentration Na₂S was injected into above solution. And it should be emphasized that this process is critical. And the solution of Na₂S cannot be added quickly, in about 30 minutes. After continuously stirring 10 h, the mixed solution was then transferred into Teflon-lined stainless-steel autoclave and heated at 110°C for 2 h. After filtration and washing with deionized water several times, the resulting precipitant was dried using a vacuum freeze dryer overnight to finally obtain NiS@L-cysteine nanosheets.

(2) Preparation of Ni₂P/NiS@PCOS: 0.1 g NiS@L-cysteine nanosheets and sodium hypophosphite hydrate powder were respectively placed in the both end of a rectangle porcelain boat. And then the porcelain boat was tightly packaged with aluminum foil and put into a tube furnace. The porcelain boat must be tightly wrapped in aluminum foil, while ensuring that sodium hypophosphate and NaS do not mix. The furnace was charged with nitrogen as the carrying gas before being turned on, which was maintained at a flow rate of 20 mL/min, and the furnace reached the desired temperature (350°C) with a constant ramp rate of 5°C/min. After 180 min reaction, the Ni₂P/NiS@PCOS heterojunction was obtained.

Supplementary note 2: role of MnO₂

In the process of reaction, the crystal structure and microstructure of Ni₂P/NiS@PCOS were found to be severely etched after the reaction in the photocatalytic overall water splitting reaction when without any sacrificial agent or MnO₂. This is because H₂O₂ can etch the catalyst (ACS Catal. 2017, 7, 1637–1645). And the addition of MnO₂ to the reaction system is an effective way to quickly remove generated H₂O₂, thereby preventing photocatalyst poisoning (J. Propul. Power.1996, 12, 574-579). During this process, the formed H₂O₂ can be removed in time by adding MnO₂, which prevents the H₂O₂ from etching the catalyst. But once the etching has taken place, it cannot be recovered. So, the main function of MnO₂ is to decompose the H₂O₂ produced by photocatalysis into O₂, and the second function is to eliminate the formed H₂O₂ in a timely manner, thus preventing the etching of the catalyst by H₂O₂. So, when MnO₂ is added to the reaction system, Ni₂P/NiS@PCOS shows excellent and stable photocatalytic hydrogen and oxygen production rates (Figure 3a-b). We have also completed characterizing and analyzing the samples of Ni₂P/NiS@PCOS samples without and with adding MnO₂ after reaction by XRD tests. When adding MnO₂ in the system of reaction, the Ni₂P/NiS@PCOS sample exhibited better structural stability, which indicated that the H₂O₂ was decomposed in time by MnO₂, so the sample of Ni₂P/NiS@PCOS could not be etched (Supplementary Figure 16 a-f).

Supplementary notes 3: further discussion on the average lifetime

In the test of time-resolved fluorescence decay, the average lifetime (τ) of $\text{Ni}_2\text{P}@PCOS$ is shorter than that of Ni_2P , but this result does not represent a contradiction with our design ideas. Due to the good charge separation effect, the fluorescence lifetime is long. But the fluorescence lifetime does not mean that the charge separation effect is ever excellent. The length of the electron lifetime is affected by various factors such as the charge separation efficiency, the electron mobility path length, and the interface reaction energy barrier. The reason why the average lifetime (τ) of $\text{Ni}_2\text{P}@PCOS$ is shorter than that of Ni_2P may be attributed to the photostimulated electron migration path of $\text{Ni}_2\text{P}@PCOS$ is shorter than that of Ni_2P . Thus, the average lifetime (τ) of $\text{Ni}_2\text{P}@PCOS$ is shorter than Ni_2P . In order to adequately demonstrate that the charge separation effect of Ni_2P with NiS and PCOS , we further investigated the photocurrent test and the photoluminescence spectroscopy test. Figure 4b showed the $\text{Ni}_2\text{P}/\text{NiS}@PCOS$ presented the strongest photocurrent response signals (about $155 \mu\text{A cm}^{-2}$), which was about 8 and 31 times as high as that of the pure NiS (about $20.0 \mu\text{A cm}^{-2}$) and $\text{Ni}_2\text{P}@PCOS$ (about $5 \mu\text{A cm}^{-2}$), respectively. Furthermore, the diameter of the Nyquist circle of $\text{Ni}_2\text{P}/\text{NiS}@PCOS$ (Supplementary Figure 25) is much smaller than that of pure NiS and $\text{Ni}_2\text{P}@PCOS$. This proved that the $\text{Ni}_2\text{P}/\text{NiS}@PCOS$ has ultrahigh electron conductivity and mobility. In addition, as shown in Supplementary Figure 26, the PL spectrum indicates that Ni_2P exhibits strong emission peaks, suggesting that electron-hole pairs recombine easily. But, in the PL spectrum of $\text{Ni}_2\text{P}@PCOS$, $\text{Ni}_2\text{P}/\text{NiS}$ and $\text{Ni}_2\text{P}/\text{NiS}@PCOS$, a weaker emission peak at the same position can be observed, indicating that the addition of cocatalyst significantly inhibits the recombination of electron-hole pairs. In Figure 4d, the time-resolved fluorescence decay spectra also further illustrated the advantages of coating PCOS and NiS on the surface of Ni_2P for the improvement of charge separation. the average lifetime (τ) for $\text{Ni}_2\text{P}/\text{NiS}@PCOS$ sample are increased, indicating that Ni_2P with double co-catalyst of NiS and PCOS possesses much longer lifetimes of photogenerated charge carriers than those of Ni_2P , $\text{Ni}_2\text{P}@PCOS$ and $\text{Ni}_2\text{P}/\text{NiS}$.

REFERENCES

- (1) Domen K, Naito S, Soma M, et al. Photocatalytic decomposition of water vapour on an NiO–SrTiO₃ catalyst[J]. *J. C. S. Chem. Comm.* **12**, 543-544 (1980).
- (2) Kudo A, Tanaka A, Domen K, et al. Photocatalytic decomposition of water over NiO K₄Nb₆O₁₇ catalyst[J]. *J. Catal.* **111**, 67-76 (1988).
- (3) Abe R, Sayama K, Domen K, et al. A new type of water splitting system composed of two different TiO₂ photocatalysts (anatase, rutile) and a IO₃⁻/I⁻ shuttle redox mediator[J]. *Chem. Phys. Lett.* **344**, 339-344 (2001).
- (4) Maeda K, Domen K. New non-oxide photocatalysts designed for overall water splitting under visible light[J]. *J. Phys. Chem. C.* **111**, 7851-7861 (2007).
- (5) Maeda K, Lu D, Domen K. Direct water splitting into hydrogen and oxygen under visible light by using modified TaON photocatalysts with d0 electronic configuration[J]. *Chem. Eur. J.* **19**, 4986-4991 (2013).
- (6) Liu J, Liu Y, Liu N, et al. Metal-free efficient photocatalyst for stable visible water splitting via a two-electron pathway[J]. *Science* **347**, 970-974 (2015).
- (7) Kibria M G, Chowdhury F A, Zhao S, et al. Visible light-driven efficient overall water splitting using p-type metal-nitride nanowire arrays[J]. *Nat. Commun.* **6**, 1-8 (2015).
- (8) Julia Soldat, G. Wilma Busser, Martin Muhler, and Michael Wark* Cr₂O₃ Nanoparticles on Ba₅Ta₄O₁₅ as a Noble-Metal-Free Oxygen Evolution Co-Catalyst for Photocatalytic Overall Water Splitting. *ChemCatChem.* **8**, 153–156 (2016).
- (9) Ham Y, Hisatomi T, Goto Y, et al. Flux-mediated doping of SrTiO₃ photocatalysts for efficient overall water splitting[J]. *J. Mater. Chem. A.* **4**, 3027-3033 (2016).
- (10) Yu Lei Wang, Ting Nie, Yu Hang Li, Xue Lu Wang, Li Rong Zheng, Ai Ping Chen, Xue Qing Gong, and Hua Gui Yang*. Black Tungsten Nitride as a Metallic Photocatalyst for Overall Water Splitting Operable at up to 765 nm. *Angew. Chem. Int. Ed.* **56**, 7430–7434 (2017).
- (11) Wang L, Zheng X, Chen L, et al. Van der Waals Heterostructures Comprised of Ultrathin Polymer Nanosheets for Efficient Z-Scheme Overall Water Splitting[J]. *Angew. Chem. Int. Ed.* **130**, 3512-3516 (2018).
- (12) Zheng Wang, Kazunari Domen et al. Overall water splitting by Ta₃N₅ nanorod single crystals grown on the edges of KTaO₃ particles. *Nat. Catal.* **1**, 756-763 (2018).
- (13) Wolff C M, Frischmann P D, Schulze M, et al. All-in-one visible-light-driven water splitting by combining nanoparticulate and molecular co-catalysts on CdS nanorods[J]. *Nat. Energy.* **3**, 862-869 (2018).
- (14) Pan Z, Zhang G, Wang X. Polymeric carbon nitride/reduced graphene oxide/Fe₂O₃: all-solid-state Z-scheme system for photocatalytic overall water splitting[J]. *Angew. Chem. Int. Ed.* **131**, 7176-7180 (2019).
- (15) Zhang J, Bai T, Huang H, et al. Metal–Organic–Framework–Based Photocatalysts Optimized by Spatially Separated Cocatalysts for Overall Water Splitting[J]. *Adv. Mater.* **32**, 2004747 (2020).
- (16) Chen S, Vequizo J J M, Pan Z, et al. Surface Modifications of (ZnSe)_{0.5} (CuGa_{2.5}Se_{4.25})_{0.5} to Promote Photocatalytic Z-Scheme Overall Water Splitting[J]. *J. Am. Chem. Soc.* **143**, 10633-10641 (2021).

- (17) Xiaoqi Zheng, et al, High Carrier Separation Efficiency in Morphology-Controlled BiOBr/C Schottky Junctions for Photocatalytic Overall Water Splitting, *ACS Nano*. **15**, 13209-13219 (2021).
- (18) Zhao D, Wang Y, Dong C L, et al. Boron-doped nitrogen-deficient carbon nitride-based Z-scheme heterostructures for photocatalytic overall water splitting[J]. *Nat. Energy*, **6**, 388-397 (2021).
- (19) Farahi E, Memarian N. Nanostructured nickel phosphide as an efficient photocatalyst: effect of phase on physical properties and dye degradation[J]. *Chem. Phys. Lett.*, **730**: 478-484 (2019).
- (20) Zhao S, Xu J, Yu H, et al. RGO boosts band gap regulates for constructing Ni₂P/RGO/MoO₂ Z-scheme heterojunction to achieve high efficiency photocatalytic H₂ evolution[J]. *Catal. Lett.*, **149**, 3012-3026 (2019).
- (21) Lou Y, He J, Liu G, et al. Efficient hydrogen evolution from the hydrolysis of ammonia borane using bilateral-like WO_{3-x} nanorods coupled with Ni₂P nanoparticles[J]. *Chem. Commun.*, **54**, 6188-6191 (2018).
- (22) Li Y, Jin Z, Wang H, et al. Effect of electron-hole separation in MoO₃@Ni₂P hybrid nanocomposite as highly efficient metal-free photocatalyst for H₂ production[J]. *J. Colloid. Interf. Sci.*, **537**, 629-639 (2019).
- (23) Liu Y, Zhao Y, Wu Q, et al. Charge storage of carbon dot enhances photo-production of H₂ and H₂O₂ over Ni₂P/carbon dot catalyst under normal pressure[J]. *Chem. Eng. J.*, **409**, 128184 (2021).
- (24) Li S H, Zhang N, Xie X, et al. Stress-Transfer-Induced In Situ Formation of Ultrathin Nickel Phosphide Nanosheets for Efficient Hydrogen Evolution[J]. *Angew. Chem. Int. Edit.*, **130**, 13266-13269 (2018).
- (25) Tian B, Li Z, Zhen W, et al. Uniformly sized (112) facet Co₂P on graphene for highly effective photocatalytic hydrogen evolution[J]. *J. Phys. Chem. C.*, **120**, 6409-6415 (2016).
- (26) Li N, Ding Y, Wu J, et al. Efficient, full spectrum-driven H₂ evolution Z-scheme Co₂P/CdS photocatalysts with Co–S bonds[J]. *ACS Appl. Mater. Inter.*, **11**, 22297-22306 (2019).
- (27) Xue Z H, Su H, Yu Q Y, et al. Janus Co/CoP nanoparticles as efficient Mott–Schottky electrocatalysts for overall water splitting in wide pH range[J]. *Adv. Energy. Mater.*, **7**, 1602355 (2017).